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INSERTION OF DATA MINING TECHNIQUES WITH FEATURE SELECTION TO PREDICT SHARE PRICE ON STOCK EXCHANGE DATA

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DATA MINING TECHNIQUES WITH FEATURE SELECTION:
A PREDICTIVE MODELING STUDY ON STOCK EXCHANGE DATA

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ABSTRACT

This study examines both the combination of data mining with feature selection in order to obtain improved stock price forecasts using machine learning approaches. The research concentrates on the UK stock exchange where the past data was retrieved, pre-processed and analysed to generate accuracy and uniformity of results. EDA has shown significant structures in the trends in stocks and statistical testing; correlation and recursive feature elimination methods were used in the feature selection process to select variables that significantly influence making stock prices change.

Three regression models: Linear Regression, Support Vector Regression (SVR), and Random Forest Regression, were created and assessed on the basis of several evaluation metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and R^2 score. Linear Regression was the baseline employed and this provided a low accuracy as it could not take into consideration the complex non-linear correlation. VR enhanced this by modelling more complex patterns but ended up being susceptible to parameter optimization. SVR performed the best across the board consistently underperforming the other models with the lowest RMSE, MAE values, and the highest R 2 score. Its population-based methodology lowered over-fitting and increased predictive performance robustness making it the model with the best performance in this study.

The paper articulates that feature selection combined with ensemble methods has a substantial outcome in the improvement of the accuracy in stock price forecasting and that the implementation of Random Forest into an interactive dashboard constitutes practical value when used by investors and analysts.

Keywords:

- EDA: Exploratory Data Analysis
- GUI: Graphical User Interface
- MSE: Mean Squared Error
- RMSE: Root Mean Squared Error
- MAE: Mean Absolute Error
- R^2: R-squared
- ML: Machine Learning
- LR: Linear Regression
- SVR: Support Vector Regression
- RFR: Random Forest Regression

Preface

This is a project that investigates the combination of data mining and feature selection methods to predict the stock market prices of the UK by the use of machine learning. It embodies the academic and the practical skills in data analysis required at the MSc stage and can be referred to as the National Qualifications Framework in that understanding and solving complex problems, critical judgement of practical data and creation of an interactive deployment interface are demonstrated. It is a work that negates the gap between the theory and practice of financial analytics and demonstrates how a smart system can be used to improve the process of making stock market predictions and decisions by both an analyst and an investor.

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Chapter 1: Introduction

1.1 Introduction

Volatility and the randomness associated with stock markets make it difficult to make accurate forecasts on shares. This study seeks to study the process of applying data mining combined with feature selection techniques in an attempt to improve stock price forecasting based on historical data of the stock exchange. The proposed approach expects to design a system to adapt the strengths of the machine learning models to allow it to focus on what it believes are the key features that lead to change in the prices, hence enhancing the predictive capability. Since this methodology is based on time series analysis, it takes into account temporal relations and market trends historically existing in data (Wang and Wang, 2021). Feature selection is an optimal choice to exclude all unnecessary information, which contributes to the improvement of numerical model training. Some of the predictive models used in data mining include the regression method, classification technique, and clustering model. Moreover, the use of the predictive models is encouraged to be implemented on a web-based platform for easy access and real time analytics. This research also adopts an extensive review of related studies, description of methodology, experimentation, and deployment approaches to advance knowledge on the application of predictive analytics in the financial markets. The results should help investors and analysts make better decisions by offering correct and timely predictions of the stock prices.

1.2 Background

Forecasting stock prices has remained an interesting and difficult task for a long time since the behaviour of the stock market is unpredictable, non-linear, and highly dynamic in most cases. Most of the basic statistical techniques fail to provide a comprehensive analysis of the data chiefly composed of historical stock information regarding various shares and securities (Fathali, Kodia, and Ben Said, 2022). The incremental use of large data sets, along with the techniques of data mining and machine learning, can extend its potential to understand the definitive values and the enhanced predictive capability. Feature selection streamlines this process in the same way as moving from a raw explanation to featuring extraction, but the idea here is to find out which variables have the greatest impact on the stock price to improve the accuracy of the model and also to reduce computational demands.

These improvements in computing capabilities and data access enabled the evaluation of large databases, and identification of concealed patterns, and the ability to make highly accurate predictions of future stock prices. Various methods, including the regression and decision trees, support vector machines, and neural networks, have been found to have a strong bearing in financial forecasting (Viéitez, Santos, and Naranjo, 2024). Large dataset can be managed by different types of data mining techniques. However, this specific area is continuously evolving and it changes whenever new technology comes. As a result, different types of issues also occurred due to the implementation of ML on such learge set of data.

Different methods, among them the decision trees, support vector machines, and neural networks, and the regression have been proven to play a very significant role in forecasting in the finance industry (Viéitez, Santos, and Naranjo, 2024). Furthermore, time series analysis offers valuable toolkits for examining trends in the variables over time, in the stock data, both for the long and short term. Employment of web-based approaches has also emerged as prominent when it involves the implementation of models of ‘‘black box ’’ used in working contexts for analysis and enhanced access by analysts and by investors. In association with feature selection, this paper discusses the failures of the traditional method to the problem and provides a more efficient and accurate method based on the data mining model.

Financial markets already are complex; external stimuli, including economic indicators, geopolitical events, market sentiment, and regulatory changes add another dimension to this complexity, in which the multidimensional relationships are not captured well using traditional forecasting models. Data-driven modern solutions to such challenges include the integration of a wide variety of data, such as technical indicators, fundamental analysis indicators, and alternative data, including social media sentiment and news analysis (Wang and Wang, 2021). Blending a combination of ensemble and hybrid modeling techniques has become an attractive approach towards harnessing the power of various algorithms and downplaying the deficiency of individual models. Additionally, the increasing interest in having the capability of issuing real-time predictions has led to the need to have effective computational frameworks capable of working with streaming data to generate timely predictions to fit the requirements of dynamic trading conditions, and as such the need to enhance complex predictive models is becoming a necessity to aid in the current financial decision-making process.

1.3 Objectives

- ➢ To gather and prepare stock exchange data from open-source resources to ensure data quality and consistency for stock price forecasting using the Python programming language.

- To apply data mining techniques to analyse historical stock price trends and identify relevant patterns.
- To implement feature selection methods to determine the most significant factors influencing stock price movements.
- To build a predictive modelling approach using machine learning algorithms to forecast stock prices with improved accuracy.
- To evaluate the model's performance using statistical metrics to validate its predictive capability with data-driven insights.
- To deploy the model in a web interface for real-time stock price prediction and user accessibility.

1.4 Research questions

Research Question 1: Can feature selection techniques be used to establish the factors that have the greatest impacts on stock price fluctuations in the UK stock market?

Research Question 2: How does data mining technique impact the performance of the machine learning models in prediction of stock price if data pre-processing is done by handling missing values and normalizing it?

1.5 Signification of the Project

The purpose of undertaking such work is due to the contribution this project holds towards improving the efficiency and specificity of stock price prediction, where data mining and feature selection aspects are applied. The flaw with the traditional methods of forecasting is that they are usually overfitting, have low generalization ability, and are computationally costly due to the continually increasing dimensionalities of the data being analyzed. As a result of this, this project successfully identifies the most influential features that can aid in making proper predictions, thus making it quicker and efficient.

There will be great usefulness in constructing an accurate model of stock price prediction since it is beneficial for investors, financial analysts, and policy makers to make effective decisions and minimize risks (Jing *et al.* 2021). Also, having web-based deployment means that these predictive models are operational and available for use at any time without delay thereby eliminating the gap existing between sophisticated analyses and decision-making processes in finances. This research is relevant to the enhancement of the predictive analytics for the stock markets, especially in the finance departments, and pulls into focus the integration of the machine learning with feature selection methods.

In addition to personal monetary gains, the history of the project guarantees overall stability and transparency in the market due to systematic methods of the price discovery and trend analysis. The methodological framework created below already forms a template that could be reproduced when dealing with varying financial markets, asset classes, and economic contexts, thus facilitating the standardization of the quantitative finance research methodology (Mu *et al.* 2023). Also, merging easy to use interfaces with powerful analysis engines democratizes access to powerful financial modelling software by empowering small investment firms and retail investors to access institutional quality analytics.

1.6 Project Outline

The paper has been divided into six major sections that follow each other to establish a coherent flow, as explored in detail in this outline. Related literature discusses literature assessment on stock rate prediction, data mining, and selection of features. It outlines the steps of data gathering and cleaning, the selection of features, and the employed methods of machine learning. Data Analysis encompasses looking at the previous records of stock and assessing the performance of a given model. Between Findings and Discussion, the former provides the results while the latter aims at pattern recognition and model comparison (Sonkavde *et al.* 2023).. The conclusion highlights the findings of the study, focusing on the success of the presented approach. At last, Recommendation identifies recommendations/ proposals for future application of data mining in an attempt to increase the accuracy of Stock price prediction.

Chapter 2: Literature Review/Background Research

2.1 Introduction

Earlier research into the forecasting of historical stock prices shall be presented in the chapter for the evaluation of literature. This chapter identifies earlier research that resorts to data mining algorithm-based approaches for approximating stock prices, especially for the United Kingdom. The analysis of this chapter mainly concentrates heavily on efficient approaches to gather different information from internet and from different scholarly articles (Kumbure et al., 2022). This chapter also evaluates method that earlier researcher used in utilizing historic knowledge about shares, time series statistical models, and machine learning for stock prediction.

2.2 Stock Price Prediction Using Data Mining Techniques

As per the view of Alshammari et al., (2022), businesses as well as shareholders need data mining technologies to predict stock values and make informed decisions. Regression techniques permit analysts analyse massive data sets and find undetectable trends. These methods include SVM, decision trees, and random forests. These tactics may illuminate stock price factors, increasing their chances of success. This has affected energy costs, foreign currency exchange rates, and socioeconomic indexes. Using these components, price variations may be predicted. Data mining methods like classification algorithms may accurately estimate the return on investment. Using existing data and relevant parameters, these approaches have predicted asset value changes multiple times (Alshammari et al., 2022).

As per the view of Rouf et al., (2021), Data extraction is important because it can examine large datasets, find concealed links, and boost projections. By offering more options, these solutions boost effectiveness and reduce uncertainty. In unpredictable markets, excellent anticipating is crucial. Analysis of data is crucial in comprehending the stock exchange factors. This makes these approaches vital for predicting financial outcomes. Market actors may better predict trends, adjust strategies, and improve their business results by using data analysis in stock price projections.

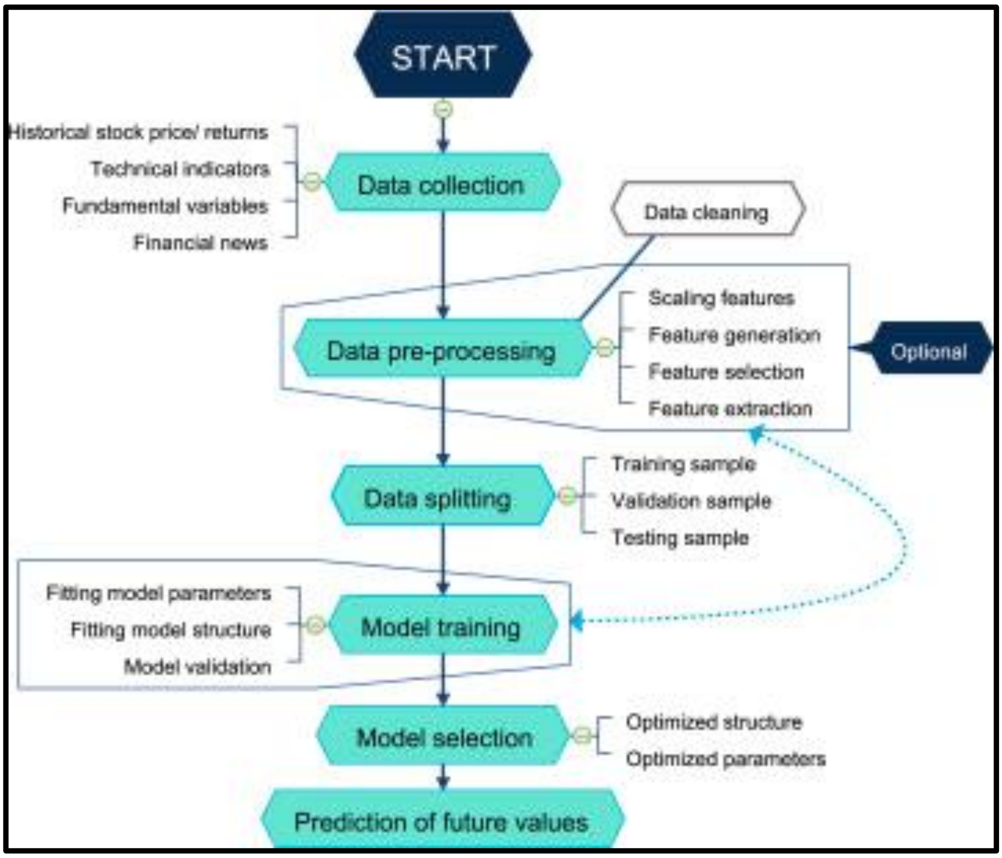

Figure 2.2.1: Data mining process to forecast historical stock exchange data

(Source: Kumbure et al., 2022)

The figure displays an organised workflow of data mining based on stock price estimation. These involve data collection steps like share market prices, indicators, news, and on the other hand, preprocessing processes such as cleaning, and feature engineering. There are training, validation and testing sets for storing data. Parameter tuning and optimization to improve the models for predicting future stock values are included in the process of Model training (Kumbure et al., 2022). In order to predict future price movements of UK stock, data mining techniques are used to analyze historical market data as well as economic data and sentiment. For example, it includes predicting the trends of FTSE 100 through the use of macroeconomic variables and analyzing the sentiment of retail investors from social networks to give predictions of volatility (Kaur and Dharni, 2023). These help the investors to make decisions by identifying patterns, correlations and anomalies. However, the share market's unreliability as well as outside elements can restrict the appropriate accuracy to the extent that advanced models have to be implemented regularly.

2.3 Feature Selection Methods in Predictive Modeling for Stock Market Data

As reported by Li, (2022), picking characteristics is crucial to forecasting stock market data. In order to fulfil this purpose, it must identify the most important qualities or attributes from a broader set of inputs. Complex stock market data is common. It includes price movements in stocks, the volume of transactions, trends, and more. Feature selection reduces the number of attributes utilised. This strategy ensures that only pertinent factors are utilised to create forecasting models, preventing overfitting and unnecessary computations.

Parameters are commonly selected using filtration algorithms. These algorithms establish features based on statistical examinations or the relationship between qualities and the variable. Pamuccu and Barbu (2024) make it possible for an attribute to be utilised if it significantly affects the valuation of stocks. Wrapper methodology uses feature subsets and forecasting models like a decision tree as well as a neural network to assess them. This method makes it easier to make a personalised decision, but it might demand a lot of resources. Lasso regression constitutes an integrated feature selection method for model training. Transferring weights to attributes and automatically removing features with low prediction accuracy permits this. Lasso may reduce model features while maintaining or improving effectiveness (Li et al., 2024).

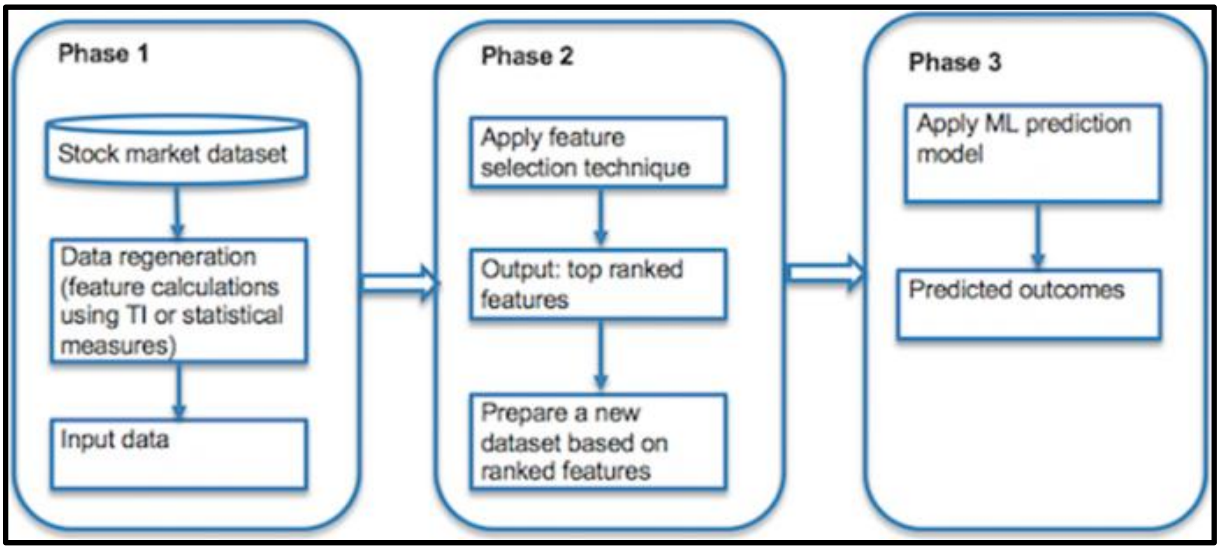

Figure 2.3.1: Application of feature selection procedure for stock market forecasting

(Source: Htun, Biehl and Petkov, 2023))

The illustration shows a three-step process for selecting stock market forecast qualities. Initial input data is produced using statistical measurements or technological characteristics. The second step analyses and refines significant characteristics using feature selection algorithms. In the third step, a machine learning algorithm can be applied to an adjusted dataset to improve stock market forecasts. The ultimate objective of choosing characteristics in predicting stock markets is to diminish data size and eliminate unneeded variation while enhancing the framework's capacity to adapt to fresh data, which improves predictability.

2.4 Historical Stock Price Trend Analysis and Pattern Recognition of United Kingdom Stock Exchange Data

As opined by Staffini, (2022), examining past price movement is a type of historical stock price trend analysis as well as pattern recognition in the data of the United Kingdom Stock Exchange to find patterns that can predict the future behaviour of the market. Statistical and computational methods of such an analysis have been used predominantly, for example, technical indicators like descriptive statistics, candlestick patterns, and recognizable chart patterns such as head-and-shoulders or double tops. These techniques have been shown through research based on historical UK stock market data to be able to provide investors with some insights into possible future price movements. For instance, empirical tests on technical patterns that portfolio shifts are predicted by specific formations, and hence may serve as profitable trading signals (Vuong et al., 2024). Nevertheless, more recently, machine learning algorithms like neural networks and deep learning developed are used for enhancing pattern detection from increased accuracy.

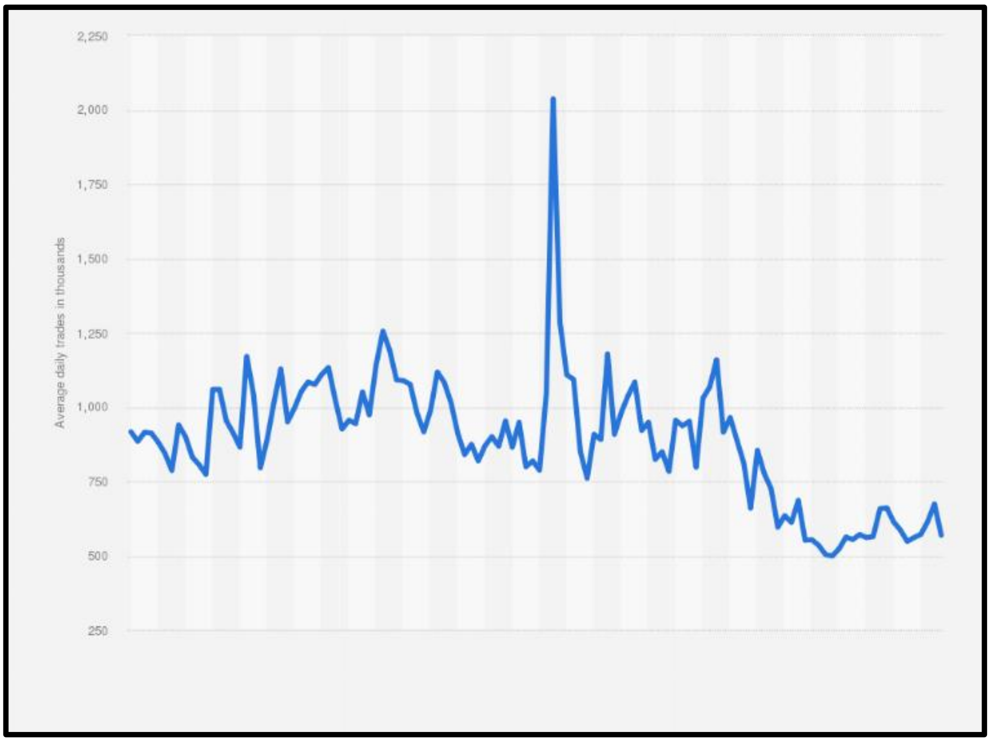

Figure 2.4.1: UK Daily Stock Exchange analysis

(Source: Statista, 2025)

The average daily trades (in thousands) on the LSE between 2020 to 2024 can be seen in the graph. It had a sharp spike in March 2020 and started to fall to 797,175 in December 2021. Up until early 2022, trades rebounded above one million, but they tapered to 568,000 by December 2024, and they dropped considerably post-2023.

Based on the extensive historical datasets to recognize the subtle & complex pattern beyond human capability, these models identify. In order to time investment decisions in the UK market, technical analysis tools like moving averages have been applied and evaluated for their effectiveness (Nikou, Mansourfar and Bagherzadeh, 2019). It is shown that these tools provide evidence for market efficiency as well as for the forecasting of stock prices. On the other hand, proponents argue that pattern-based strategies have correctly predicted anomalies in the UK market, seasonal effects and short-term price variations (Vuong et al., 2024). Therefore, historical stock trend analysis and pattern recognition have a positive promise in explaining UK market dynamics, though their success depends on the evolution of markets as well as the complexity of financial systems.

2.5 Machine Learning Algorithms for Stock Price Forecasting

As argued by Singh et al., (2019), historical data is leveraged by machine learning (ML) algorithms to predict future market behaviour of stock prices and thus have significantly advanced stock price forecasting. Linear Regression, Support Vector Machines (SVM), Random Forest, and Long Short Term Memory (LSTM), are due to their capability to cope with complex patterns and non-linear relationships in stock market data. For instance, linear Regression foresees future prices by managing on a linear correlation line among independent variables (such as the historical price, the volume) and dependent variables (future price), also measured by root mean square error (RMSE) (Bansal, Goyal and Choudhary, 2022). It has been noticed that simpler models such as Random Forests and XG Boost tend to always be performing better than simple models, and usually exceed 80% accuracy and relatively low forecasting errors (Soni, Tewari and Krishnan, 2022). On the contrary, since LSTM algorithms have sequential memory ability to retain significant past information and are thus suited to handle long-term dependencies in price movements, LSTM algorithms are best suited for volatile financial data.

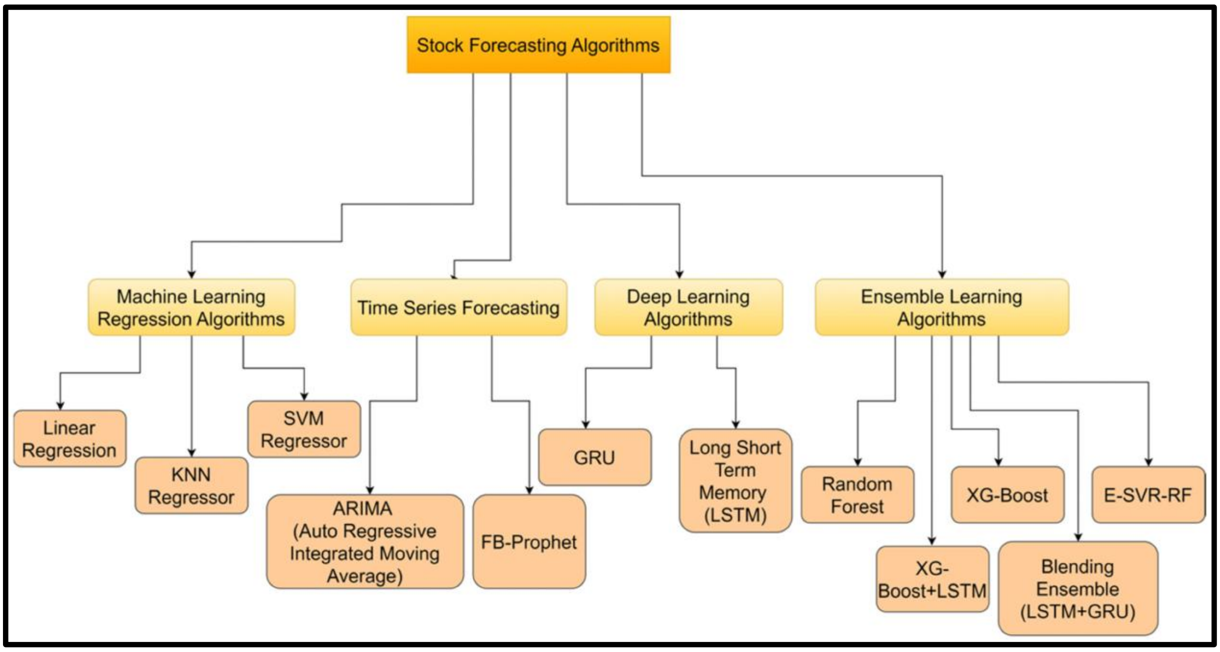

Figure 2.5.1: Deployment of machine learning algorithms for the stock market prediction

(Source: Sonkavde et al., 2023)

The diagram portrays that Machine Learning Regression (Linear Regression, KNN, SVM), Time Series (ARIMA, FR-Proplet), Deep Learning (GRU, LSTM), and Ensemble Methods (Random Forest, XG-Boost, E-SVR-RF) are the four categories of the stock forecasting algorithms. Other hybrid models like XG-Boost, LSTM and Blending Ensemble (LSTM and GRU) are mentioned as other ways to introduce better prediction accuracy by blending the strengths of many algorithms (Sonkavde et al., 2023).

The stock prices are extensively forecasted by machine learning models which include ARIMA-LS-SVM, ARMA-GARCH-NN, and the k nearest neighbours (kNNC). Stock market data has sequential patterns which can be captured by deep learning models, in this case, LSTM and GRU, which result in accuracy and less error as shown by Mean Absolute Percentage Error (MAPE) and Root Mean Square Percentage Error (RMSPE) (Lawi, Mesra and Amir, 2022). Furthermore, transfer was learning methods handle data insufficiency problems and therefore, result in better forecasting performance by integrating external data. In most cases, traditional machine learning is outperformed by deep learning-based models as these can capture complex dynamics over time.

2.6 Importance of Feature Selection and Machine Learning Techniques in Data Mining for Stock Price Prediction

The stock price prediction applies its feature selection techniques which help to achieve better model accuracy and efficiency by selecting the best variables out of an enormous dataset. Feature selection helps reduce noise, remove redundant or irrelevant data features, simplify the model, and reduce the risk of overfitting while improving the model's predictiveness. Statistical methods like moving averages (or any indicators), Relative Strength Index (RSI), and trading volume help predict more than if one included the irrelevant sentiment scores (Soni, Tewari and Krishnan, 2022). Furthermore, correlation-based filters or recursive feature elimination are used to remove factors like historical prices or economic indicators which are more influential in addition to helping foresee the stock and also aiding with decision-making in investing.

Stock price prediction is done by selecting machine learning models as they are capable of effectively detecting complicated and non-linear market patterns in the historical data, thereby enhancing forecasting accuracy and adaptability to the market, say neural network and random forest models are more adept at finding out subtle market dynamics than traditional methods. For the stock exchange, applying ML forecasting methods is essential since traditional analytical methods comparable to fundamental or technical analysis fall short when dealing with invariant, noisy and nonstationary financial data and too late or wrong predictions (Sonkavde et al., 2023). Unlike humans, ML models can outperform in such situations because they can quickly learn the subtle interactions between many indicators and will not get tired in thousands of iterations. As such, ML algorithms greatly help in stock market forecasting, and hence, they are critical components in modern financial analysis.

2.7 Real-Time Stock Price Prediction and Web Deployment

As stated by Phuoc et al., (2024), real-time stock price prediction with ML models in a web interface for the ease of the investors provides an acceptable proficiency to ease the decision-making ability of the investors. Big data are grabbed, analyzed, and processed immediately by these web-based platforms for stock market trends. ML models are integrated into a web-based graphical user interface (GUI) using frameworks such as Streamlit and Gradio (Phuoc et al., 2024; Chen et al., 2023). The application that results comes with functionality for the selection of specific companies, viewing of historical data, and access to forecast graphs, for enabled, more informed investment decision making.

Real-time stock prediction models have been developed with hybrid models, which incorporate with web-based dashboards to provide investors with accessible and interactive forecasting tools.

Models such as Linear Regression, Support Vector Regression (SVR), and Random Forest are commonly applied, with Linear Regression offering a baseline, SVR capturing non-linear dynamics while ranking indicator importance, and Random Forest handling noisy data (Akhtar et al., 2022; Fathali et al., 2022; Bansal et al., 2022; Pashankar et al., 2024). To enhance prediction accuracy, feature selection methods such as correlation-based filtering and recursive feature elimination (RFE) are employed to retain the most relevant stock indicators (Htun et al., 2023; Büyükkeçeci and Okur, 2023). Deployment is achieved using frameworks like Streamlit and Gradio, which allow real-time interaction through functionalities such as company selection, visualization of historical data, and forecast graphs (Phuoc et al., 2024; Chen et al., 2023). By combining robust ML models, feature selection, and web deployment, these systems transform predictive analytics into practical tools that support data-driven investment decisions (Kumbure et al., 2022; Sonkavde et al., 2023). These models try to improve the accuracy of the stock price forecasts by analyzing the sentiment and the information extracted. Web interfaces present these predictions, allowing users to have an overall picture of the movement of the market in response to both quantitative and public sentiment information.

2.8 Challenges in Stock Price Prediction Using Data Mining Techniques

Financial markets are so complex that the use of data mining techniques to predict stock prices is so many challenges. There is one major obstacle to breaking the dynamics of stock data: these are non-stationary and thus the statistical properties vary with time which prevents forging simple common predictive models. The market is also subject to a list of other factors that influence it in chaotic and noisy data, which blocks the view of recognizable patterns.

Further complicating the analysis is the fact that this financial data comes in a vast volume that needs to be extracted to avoid introducing noise into its analysis. In addition, the existence of market anomalies and irrational behaviours of investors can cause the price movements to be sudden and unpredictable and will contradict the prevalent trend. For instance, geopolitical occurrences or abrupt changes in economic policies cause market volatility and make data-driven models unreliable.

Additionally, the quick pace at which financial instruments and market strategies evolve implies that models may become obsolete very fast, requiring ongoing updating and validation. Overfitting is another hazard, as models may capture the fitted data very well but are poor predictors for new data. Solving these issues involves combining heterogeneous sources of information, e.g., financial news and sentiment on social media, to get a clearer understanding of market behaviour. This integration, however, poses new issues such as poor data quality as well as the requirement for advanced natural language processing. Overall, the multidimensionality as well as the dynamic nature of the financial market makes stock price prediction a challenging task.

2.9 Critical Analysis of Existing Studies

Reference Number	Results	Impacts	Limitations
(Alshammari et al., 2022)	Applied big data mining techniques to predict Kuwait's stock market, achieving approximately 50% accuracy.	Introduced novel variables and techniques for stock market prediction in the GCC region, enhancing decision-making processes.	Limited to stock market; data collection affected by COVID-19 pandemic.
(Bansal, Goyal and Choudhary, 2022)	Developed machine learning models that accurately forecast stock market movements, demonstrating superior performance over conventional statistical methods.	Provides investors with reliable tools for market trend analysis.	Models may require extensive computational resources; performance varies with market conditions.
(Kalra et al., 2024)	Introduced a hybrid model combining multiple machine learning techniques, achieving efficient and accurate forecasting of real-time stock market indices.	Enhances real-time forecasting capabilities, beneficial for active traders and investors.	Model complexity may limit real-time application feasibility; requires continuous data updates.
(Kaur and Dharni,	Proposed a hybrid	Bridges technical and	Integration

2023)	data mining approach integrating technical and fundamental analyses, resulting in improved stock price prediction accuracy.	fundamental analysis, offering a holistic prediction model for investors.	complexity may pose challenges; effectiveness depends on data quality.
(Kumbure et al., 2022)	Neural networks dominate stock prediction (39/138 studies), with technical indicators as key variables (62%).	Guides future research toward effective ML techniques.	Limited to 2000–2019 data; lacks global market diversity.
(Lawi, Mesra and Amir, 2022)	LSTM and GRU models achieved 88-98% accuracy in stock price prediction, with GRU outperforming LSTM in most cases.	Enhances forecasting precision for investors, aiding decision-making in volatile markets.	Limited to four NasdaqGS companies; may not generalize to other stocks or markets.
(Li, 2022)	LowestPrice absence improves prediction accuracy (R^2=0.971), while ClosePrice/VWAP absence harms it most.	Helps reduce input dimensionality without sacrificing accuracy.	No testing on multi-layer LSTM or optimal feature combinations.
(Li et al., 2024)	SGP-LSTM model achieved 16.38% Excess Return, outperforming other	Enhanced predictive accuracy and portfolio performance,	Negative accumulated alpha for all models, highlighting biases in

	models.	showing superiority in stock selection and forecasting.	prediction outcomes despite improvements.
(Pabuccu and Barbu, 2024)	Compared deep learning algorithms with traditional machine learning algorithms for stock price prediction.	Provides insights into the relative performance of deep learning versus traditional machine learning algorithms in stock price prediction.	Does not explore hybrid models or advanced feature selection techniques; may not reflect current advancements in the field.
(Phuoc et al., 2024)	LSTM achieved >93% accuracy for most stocks, with PNJ at 97.7%, but lower for NVL (78.9%).	Useful for investors and regulators in trend prediction.	Short test period (Jan–Apr 2021), variability across stocks.
(Rouf et al., 2021)	Machine learning techniques, including ANN, SVM, and hybrid models, demonstrate varying levels of accuracy in stock market prediction.	Increased prediction efficiency, better decision-making, and improved market forecasting.	High complexity, overfitting risk, noisy data, volatility challenges, and dependency on accurate data sources.
(Shi et al., 2022)	The proposed model (AttCLX) achieves the lowest MAE, RMSE, and MAPE, and the highest R2 value compared to	Improves prediction accuracy for stock prices using hybrid models, with better generalization.	Computationally expensive and dependent on large training data, with limited scalability for real-time prediction.

	other methods.		
(Singh et al., 2019)	Applied SVM, RF, NB, LSTM, and ANN models to predict stock prices, analyzing NASDAQ, NYSE, FTSE, and Nikkei indices.	Demonstrated that combining traditional ML models with sentiment analysis enhances prediction accuracy.	Models may face overfitting due to feature selection challenges and data dependency.
(Soni, Tewari and Krishnan, 2022)	Reviewed ML and DL techniques, including SVM, RF, NB, LSTM, and ANN, for stock price prediction.	Provided a comprehensive analysis of various ML and DL approaches, highlighting their strengths and challenges in stock price prediction.	Limited by the availability and quality of financial data, affecting model performance.
(Sonkavde et al., 2023)	Analyzed various ML and DL models for stock price forecasting, discussing performance metrics and implications.	Offered insights into the effectiveness of different models, aiding investors and analysts in model selection.	Variability in model performance due to market volatility and external factors.
(Staffini, 2022)	Proposed a GAN-based model for stock price forecasting, demonstrating its potential in capturing complex patterns.	Introduced innovative use of GANs in financial forecasting, potentially improving prediction accuracy.	GANs require large datasets and are sensitive to hyperparameter choices, posing implementation challenges.

(Vuong et al., 2024)	Conducted a bibliometric analysis of stock price forecasting methods, tracing evolution from statistical models to deep learning approaches.	Highlighted trends and shifts in forecasting methodologies, guiding future research directions.	Analysis may overlook emerging hybrid models combining statistical and machine learning techniques.

2.10 Research Gap

This chapter mainly explains that how features and data extraction can anticipate prices in the UK market. It identifies the usage of different machine learning algorithms such SVM, filter, wrapper, and embedding methods to increase prediction. Additionally, many models are based on past data, but they tend not to capture the changing nature of financial markets, which can result in models becoming outdated over time. Hybrid designs are currently being studied, they may not capture all of the complicated fluctuations in the market. Expectations vary per inventory, and political occurrences and market idiosyncrasies can lead to abrupt and unforeseen stock price movements that models cannot predict. A restriction may increase prediction reliability and efficacy when sophisticated techniques for picking features are combined with machine-learning approaches. Using data mining approaches to anticipate stock prices has made development, but updating models to market conditions in real-time is difficult.

2.11 Summary

This chapter examines how choosing features and data extraction can anticipate prices for shares in the UK market. It explores the usage of several machine learning algorithms such as support vector machines (SVM), neural networks, and filter, wrapper, and embedding methods to increase prediction accuracy. The chapter reviews related works by comparing them to each other to determine limitations and potential impacts. This chapter also covers managing extremely complex as well as chaotic accounting information challenges such as market unpredictability and irregular, data. Overcoming these problems is covered in this chapter. A requirement for mixed models, improvements in choosing characteristics, and real-time market swing responses are the study's limitations. The remaining sections of the chapter outline the shortcomings of the research.

Chapter 3: Methodology

3.1 Introduction

The problem of stock price prediction is essentially complicated because of the high number of aspects affecting the market. This chapter mainly explain how the process of implementation has been done in this project in terms of approach and code making decision, data collection and model evaluation for the scorch price detection. This chapter also discuss the different variations of machine learning model that has been used in this project and the process of data cleaning as well as model fitting procedures.

3.2 Research Design

This investigation applies a quantitative approach, using secondary data available on open platforms such as Kaggle. The main aim is to use data mining techniques together with machine learning to guess what stock prices will be in the UK market. With this approach, analysts can review objective results and develop models based on real past data, which makes it easier to notice trends and make predictions.

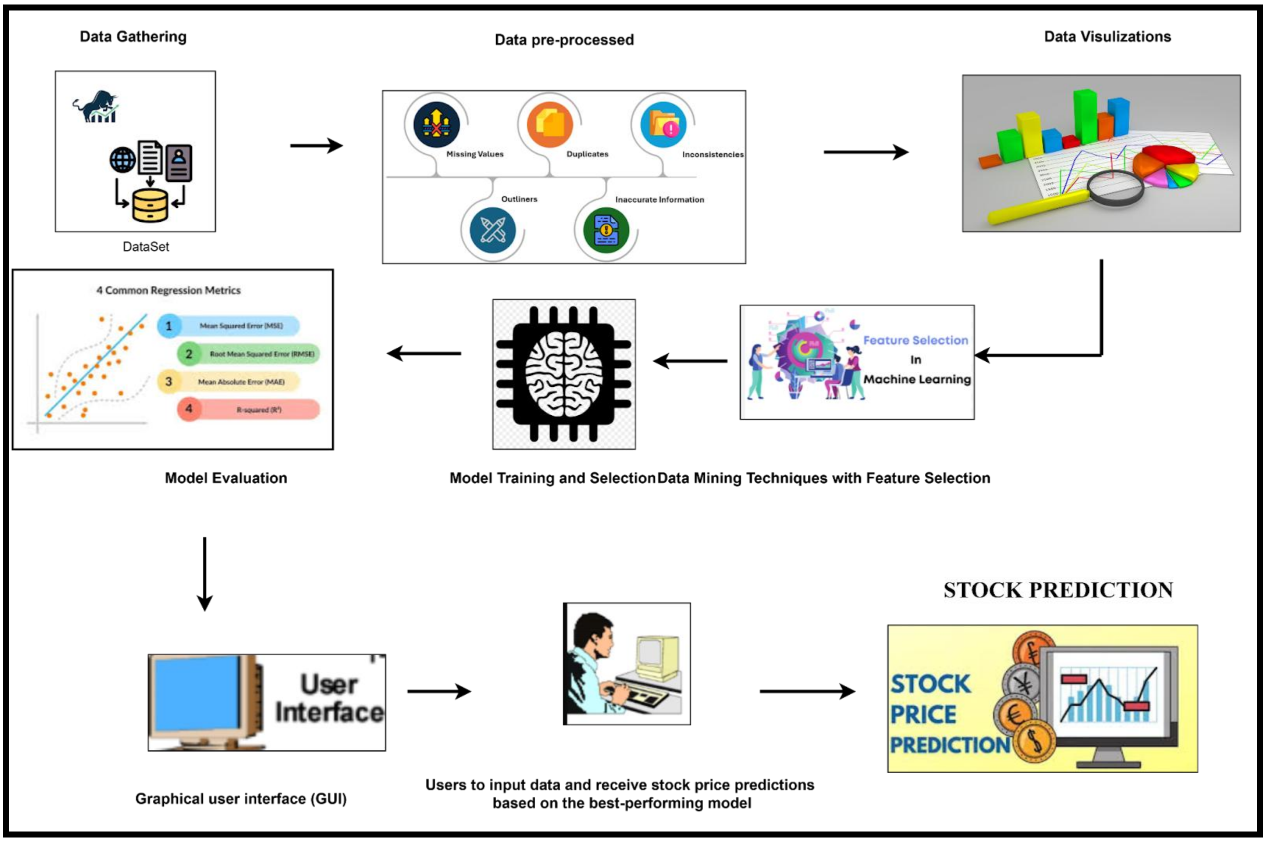

Figure 3.1: Proposed Framework to Predict Share Price on Stock Exchange Data

(Source: Self-Created)

The study used sequential phases in its research methodology, as shown in the figure. First, Data Collection is carried out to gather past stock prices from open-source systems. The next phase is Data Preprocessing, where data is cleaned, missing values are addressed, and the data is normalized to keep quality and consistency. Then, EDA is performed using graphics and statistics to spot patterns and relationships (Htun and Petkov, 2023). Feature Selection removes unnecessary variables from the dataset to make it better. In order to build the model, various Ml models such as linear regression, support vector regression are trained on the data that has been refined data. Predictive robustness of these models is measured with R-squared, mean absolute error, and root mean squared error (Bansal and Choudhary, 2022). After that, the chosen model is put into an interactive setting so users can access real-time stock predictions easily. This design method ensures that rigorously develop a useful and reliable tool for making predictions.

3.3 Data Collection and Preparation

3.3.1 Data Source

The historical data for UK company stock prices will be taken from an open-source dataset found on Kaggle, "*https://www.kaggle.com/datasets/eren2222/uk-daily-historical-stock-market-1988-2024*''. The study uses a broad data collection of UK stock market information, tracking 1,026 companies from 1988 to 2024. It includes several details, among them the date, starting price, final price for the day, daily highest and lowest prices, the total trading amount, and an adjusted closing price. The size and detail in this dataset make it reliable for thorough research and training of machine learning models, as they represent many different market trends across time. With a long history, the data can be used to thoroughly examine trends in both stock prices and the market. The broad collection of data helps to choose important information and examine its impact on stocks, securing the number one factor involved. Through using the data, the study aims to create accurate models using machine learning, giving investors and financial analysts useful tips for predicting stock prices in the UK.

3.3.2 Data Preprocessing

The initial process after collecting data is data preprocessing. To do this, it focuses on filling missing data, changing data types, handling outliers, and adjusting all features to the same level. Errors, missing pieces of data, and varying levels of data in the raw market data can hurt how your model works. For this study, preprocessing tasks are carried out using Python libraries such as Pandas, NumPy, and Scikit-learn.

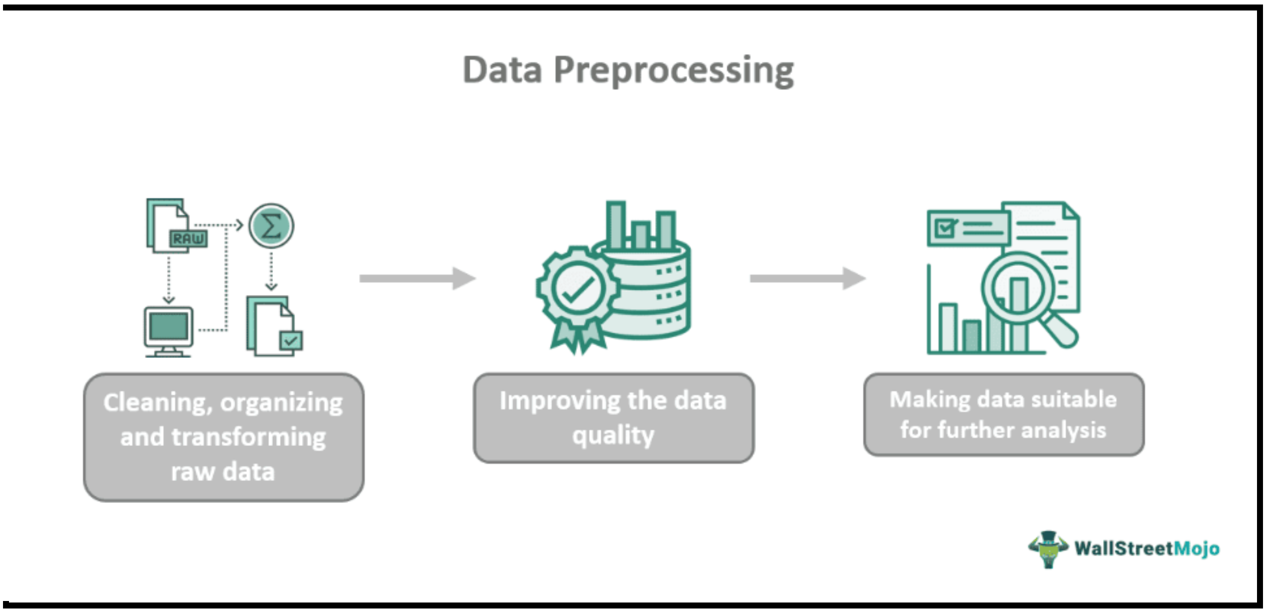

Figure 3.3.2.1: Data Preprocessing Workflow

(Source: WallStreetMojo, 2023)

Important preprocessing steps are part of a machine learning model.

- Missing Values Approach: Missing or zero data points are addressed by using replacement methods such as using the mean or median, or, if required, by deleting incomplete records to protect the quality of the whole dataset.
- Data Consistency: All dates are changed into standard datetime formats. Outliers in the values of stock prices and volumes are handled to lessen skewness and the presence of noise (Akhtar *et al.* 2022).
- Normalization: Data features are fixed to a consistent scale using one of these methods: Min-Max scaling or StandardScaler. Using this method makes the model more accurate and less influenced by large feature scales.
- The processed data is where and augmented into training (80%) and testing (20%) subsets, so the model can be trained and its performance tested without bias (Nti and Weyori, 2020).

Using this pre-processing framework which ensures the data is organized, even, and suitable for the creation and testing of models.

3.4 Exploratory Data Analysis (EDA)

Performing EDA is necessary since it helps us discover important patterns, trends, and unusual happenings in the data from the stock market. It is important to see how the data is distributed, search for patterns, and look at the way opening price, closing price, volume and adjusted close interact during EDA. Graphs and statistical summaries help find any points that look unusual, since they can negatively impact the predictive accuracy of models (Nadeem, 2024).

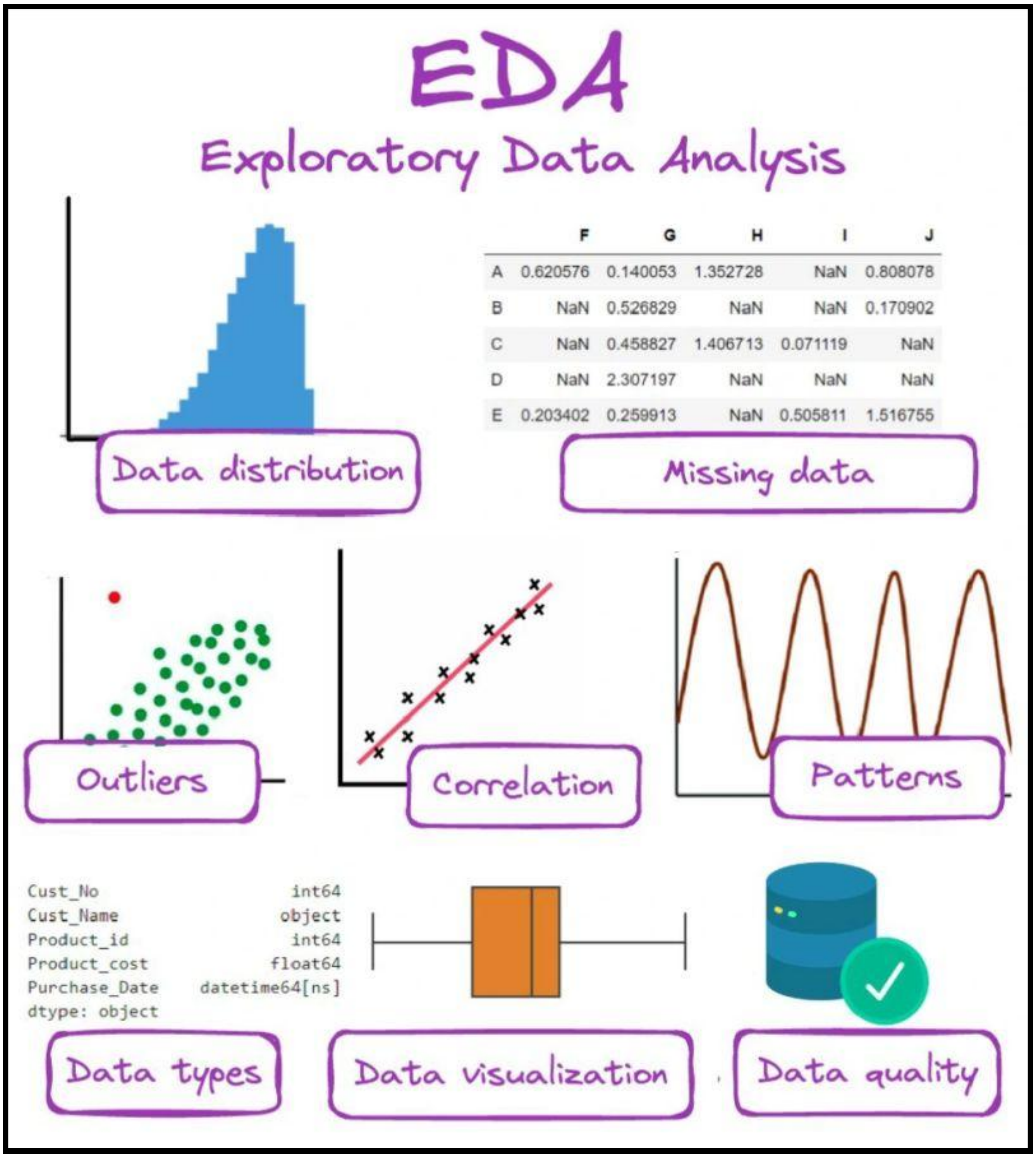

Figure 3.4.1: Example of Key Methods in EDA Process

(Source: Markov, 2023)

The figure highlights some typical EDA methods, such as plotting data, creating statistics, and looking for correlations, which support analyzing stock price trends and choosing important features for models. Performing EDA shows patterns in price variations over time, which helps pick the most useful features and the right model to use (Katya, 2023). Additionally, it proves or disproves early assumptions and makes sure the data can be used effectively for predictive analysis.

With the help of Seaborn and Matplotlib in Python, it becomes much easier to analyze the behavior of the stock market, which is essential for building useful models.

3.5 Model training methods

Training Protocol:

The training methodology of the model adopts a comprehensive cross-validation method that makes use of two strategies, namely the k-fold cross-validation and time series split, in order to assure adequate model performance evaluation. K-fold cross-validation is a way of splitting the dataset into k equal and independent parts by which the test is done k times where k-1 subsets are used to train and the remaining subset is used to test, values of the performance are averaged after k trials. The chronological structure of time split is observed in time series split where past data are utilized to prepare the tests and the rest of cases are utilized to perform the test. Grid search is the methodology used in hyperparameter optimization to explore combinations of parameters to determine optimal settings of the model. Common methods of overfitting prevention are regularization parameters, early-stopping criteria and validation monitoring that balance generalization of models to unseen data at the same time resulting in predictive accuracy on training datasets.

Validation Strategy:

Validation scheme sets aside a validation set to be used in the process of hyperparameter tuning so that the final test set is not contaminated with any biased model selection. This leaves no risk of leakage of data and preserves the integrity of performance evaluation. The methodology used is a walk forward validation procedure where the training window is renewed dynamically as more data is received and reflects the situation in real life where the models adopted by a trader have to change with the changing conditions in the market. Stability of the model acknowledges performance reliability among different timeframes such as bull markets, bear markets and in highly volatile times. Such a temporal analysis is able to confirm that the model chosen would have a stable prediction ability over the variability of market conditions with assurance on the effectiveness of deployment and viable usage in the long-term operation of dynamic variables in financial markets.

3.6 Model evaluation methods

The applied research methodology presents an exhaustive system of various evaluation measures to cover fully the performance of models in various aspects of predictive fidelity and reliability.

The multi-metric strategy guarantees a strong level of assessment that takes into account different dimensions of model performance in stock prices forecasting cases.

Accuracy Metrics:

Four of the basic regression measures are incorporated into the accuracy assessment framework in order to report the performance of predictive measures in detail. Mean Absolute Error (MAE) computes an average difference between predicted and actual stock prices, and its value is easy to understand as a measure of the accuracy of the predictions, which preserves the initial scale of the variable to be predicted. Mean Squared Error (MSE) calculates the mean of squared error so that a larger error is weighed higher than a small error thus it is sensitive to extreme outliers and large prediction deviations. Root Mean Squared Error (RMSE) is the square root of MSE, and so it expresses the error in the original scale, although it still responds strongly to large errors. Since large prediction errors have large consequences in financial applications, RMSE can be especially useful there. R-squared coefficient of determination represents the level of variance of stock prices, that is elucidated by the model and signifies the effectiveness of the model in catching underlying patterns.

Comparative Analysis:

The methodology of the comparative analysis develops statistical schemes where criteria of objective evaluation of the comparison and choice of models are built on the basis of empirical data but not subjective opinion. The methods involved in the statistical significance test between models involve using both paired t-tests and the Wilcoxon signed-rank test to define whether the difference in performance in the competing algorithms is statistically meaningful or an occurrence of chance difficulty. Confidence intervals expressed in performance measures give the boundaries of uncertainty around each measure of evaluation and can determine the reliability of results and statistical strength. The robustness of comparative analysis through cross-validation mitigates the effect of specific combinations of training/testing data on determining the choice of final models because models are judged under many different settings. Such a large comparative framework will Dashboard effective decision-making on a model selection which is optimal in terms of sound statistical evidence and applicable stakes in predicting stock prices.

3.7 Summary

This chapter has identified the entire methodology that has been followed to come up with an effective way of predicting the stock prices in the UK market. The methodical method includes acquiring data according to data sources that will allow getting high-quality data, intensive preprocessing, explorating analysis, feature selection using scientific methods, the comparison of models, and the employment strategy, which will be simple to use.

The approach is created in such a way that it is easy to repeat and is not biased as much; the results can be strongly used in practice in financial decision-making. In the following chapter, the actual usage of this methodology will be described and the empirical results of the comparative analysis of various machine learning methodologies will be provided. The systematic process that is outlined below gives a basis on how to critically evaluate effective machine learning tools in the financial forecasting tasks, which forms a wider background on data-driven processes in the stock market forecasting.

Chapter 4: Data analysis

4.1 Introduction

Analyzing data forms the foundation for both learning from it and building projections in stock market prediction. The author outlines how to study a stock market dataset step by step, covering each subject from data intake to evaluating models produced. The aim is to build a model that can use for many tasks, is still scalable and remains efficient.

4.2 Data collection

The accuracy and results of an analysis depends on how meaningfully the data are collected. Data is taken from the dataset found at
https://www.kaggle.com/datasets/eren2222/uk-daily-historical-stock-market-1988-2024.
The first section is collecting the data which can be handled through Colab’s accessible upload feature. As a result, analysts can work with their own samples, since they do not need to use samples that have been provided by others. As soon as analysts load the dataset, the system checks how many rows it contains to keep computations fast. When data includes more than 1000 observations, only 1000 rows are analyzed by random selection. By downsampling, they are able to handle the analysis and still recognize the main features of the data. With just 1000 rows or less, no data engineering is necessary. Once uploaded, the system confirms the file, shows its shape and provides a quick look at several rows to help users verify the data’s format.

4.3 Data preprocessing

4.3.1 Data Information

```
print("\nDataset Structure:")
print(df.info())

print("\nColumn Names:")
print(df.columns.tolist())

print("\nFirst 5 rows:")
print(df.head())

print("\nBasic Statistics:")
print(df.describe())
```

```
Dataset Structure:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Unnamed: 0    1000 non-null   int64
 1   Date          1000 non-null   object
 2   Ticker        1000 non-null   object
 3   Company_Name  999 non-null    object
 4   Open          1000 non-null   float64
 5   High          1000 non-null   float64
 6   Low           1000 non-null   float64
 7   Close         1000 non-null   float64
 8   Adj Close     1000 non-null   float64
 9   Volume        1000 non-null   float64
dtypes: float64(6), int64(1), object(3)
memory usage: 78.3+ KB
None

Column Names:
['Unnamed: 0', 'Date', 'Ticker', 'Company_Name', 'Open', 'High', 'Low', 'Close', 'Adj Close', 'Volume']

First 5 rows:
   Unnamed: 0        Date  Ticker                              Company_Name  \
0     1801848  2013-01-29  HPAC.L    HERMES PACIFIC INVESTMENTS PLC ORD GBP1
1     3185218  2003-07-09   REC.L                  RECORD PLC ORD GBP0.00025
2     3550126  2014-04-11   SNR.L                       SENIOR PLC GBP0.10
3     2526749  2011-10-07  MNDI.L                     MONDI PLC ORD EUR0.22
4     2482394  1998-04-27  MGAM.L  MORGAN ADVANCED MATERIALS PLC ORD GBP0.25

         Open        High         Low       Close   Adj Close    Volume
0  125.000000  125.000000  125.000000  125.000000  125.000000       0.0
1    0.750000    0.750000    0.750000    0.750000    0.294709       0.0
2  297.399994  301.600006  296.000000  299.500000  250.958542  568833.0
3  540.539917  542.519897  527.449890  533.499878  312.607971  818183.0
4  447.872192  452.318115  434.921204  437.337494  186.417862  884301.0

Basic Statistics:
         Unnamed: 0           Open           High            Low  \
count  1.000000e+03    1000.000000    1000.000000    1000.000000
mean   2.168789e+06    2772.282544    2788.520205    2739.732617
std    1.270390e+06   36516.898361   36567.538228   36376.905774
min    1.860000e+03       0.016000       0.016000       0.016000
25%    1.046249e+06      26.556200      27.667499      26.556200
50%    2.182101e+06     119.707199     123.758599     118.980301
75%    3.258433e+06     399.224640     400.449997     386.051811
max    4.399545e+06  791974.687500  791974.687500  791974.687500

              Close     Adj Close        Volume
count   1000.000000  1.000000e+03  1.000000e+03
mean    2760.601096  4.284747e+08  2.113998e+06
std    36457.693192  1.354948e+10  1.569529e+07
min        0.016000 -6.718057e+01  0.000000e+00
25%       27.462500  1.912475e+01  4.825000e+01
50%      122.057548  9.203912e+01  3.955200e+04
75%      392.999840  2.722224e+02  5.054458e+05
```

Figure 4.3.1: basic data info

(Source: Self-created)

The study lightens up the details of the dataset, highlighting the form of the data, the names of its columns, the types of data and how memory is being used (Kuroki, 2021). In the beginning, this overview provides useful knowledge about the dataset's features such as the number of entries, the number of features without missing data and the data's general integrity.

4.3.2 Data cleaning

Checking missing values

```
[8]  missing_values = df.isnull().sum()
     print(f"\nMissing Values Summary:")
     for col, missing in missing_values.items():
         if missing > 0:
             print(f"  {col}: {missing} ({missing/len(df)*100:.2f}%)")
```

```
Missing Values Summary:
  Company_Name: 1 (0.10%)
```

Figure 4.3.2.1: Checking missing value

(Source: Self-created)

The analyst goes through every column of the dataset and notes instances when values are missing (Vallejo *et al.* 2022). As a result of this systematic review, it is clear which data are missing, so analysts can better select procedures and identify possible data problems that may interfere with analysis conclusions.

Handling missing values

```
[9]  if missing_values.sum() > 0:
         print("\nHandling missing values...")

         # For numeric columns, use forward fill then backward fill
         for col in numeric_cols:
             if df[col].isnull().sum() > 0:
                 df[col] = df[col].fillna(method='ffill').fillna(method='bfill')

         # For categorical columns, use mode
         categorical_cols = df.select_dtypes(include=['object']).columns
         for col in categorical_cols:
             if df[col].isnull().sum() > 0:
                 df[col] = df[col].fillna(df[col].mode()[0] if not df[col].mode().empty else 'Unknown')

         print("✓ Missing values handled")
```

```
Handling missing values...
✓ Missing values handled
```

Figure 4.3.2.2: Handling missing value

(Source: self created)

The implementation of different approaches to fill in missing information in the data. Various approaches are used to deal with missing values, but all work to fill in the gaps and keep the data complete and numerous represent the original data group for the following study.

Removing duplicates

```
[10] initial_shape = df.shape[0]
     df = df.drop_duplicates()
     print(f"✓ Removed {initial_shape - df.shape[0]} duplicate rows")

     # Data validation for stock data
     price_cols = [col for col in df.columns if any(price_term in col.lower()
                   for price_term in ['open', 'high', 'low', 'close', 'price'])]

     if len(price_cols) >= 4:  # If we have OHLC data
         # Ensure High >= Low and other logical constraints
         high_col = [col for col in price_cols if 'high' in col.lower()]
         low_col = [col for col in price_cols if 'low' in col.lower()]

         if high_col and low_col:
             invalid_data = df[df[high_col[0]] < df[low_col[0]]]
             if len(invalid_data) > 0:
                 print(f"⚠ Found {len(invalid_data)} rows with High < Low, removing...")
                 df = df[df[high_col[0]] >= df[low_col[0]]]
```

```
✓ Removed 0 duplicate rows
```

Figure 4.3.2.3: Removing duplicates

(Source: Self-created)

To stop duplication, the analyst eliminates similar records while analyzing the data. This way, no observation is the same which makes data better and prevents possible bias during statistical and model training.

Removing negative prices

```
[11] for col in price_cols:
         negative_prices = df[df[col] < 0]
         if len(negative_prices) > 0:
             print(f"⚠ Found {len(negative_prices)} negative prices in {col}, removing...")
             df = df[df[col] >= 0]

     print(f"\nCleaned dataset shape: {df.shape}")
```

```
⚠ Found 4 negative prices in Adj_Close, removing...

Cleaned dataset shape: (996, 9)
```

Figure 4.3.2.4: Removing negative prices

(Source: Self-created)

Records that contain price values less than zero are removed from the study because they do not reflect reality. By eliminating unusual entries in the data, protect the analysis results and guarantee that all prices used for calculations and predictions are positive.

4.3.3 Exploratory data analysis

Updated statistics

```
print("2.1 BASIC STATISTICS")
print("-" * 20)


print("\nCleaned dataset statistics:")
print(df[numeric_cols].describe())
```

```
2.1 BASIC STATISTICS
--------------------

Cleaned dataset statistics:
                Open           High            Low          Close  \
count     996.000000     996.000000     996.000000     996.000000
mean     2783.390814    2799.693687    2750.710165    2771.662453
std     36589.803572   36640.540748   36449.538162   36530.482394
min         0.016000       0.016000       0.016000       0.016000
25%        28.172500      28.093750      27.844474      28.125000
50%       122.849998     125.625000     120.131649     123.985501
75%       400.000000     402.092263     387.027885     394.300003
max    791974.687500  791974.687500  791974.687500  791974.687500

          Adj_Close        Volume
count  9.960000e+02  9.960000e+02
mean   4.301954e+08  2.122488e+06
std    1.357666e+10  1.572623e+07
min    0.000000e+00  0.000000e+00
25%    1.945586e+01  6.175000e+01
50%    9.271466e+01  3.998600e+04
75%    2.725930e+02  5.064315e+05
max    4.284721e+11  3.562682e+08
```

Figure 4.3.3.1: Updated statistics

(Source: Self-created)

An analyst displays extensive descriptive statistics, identifying the measures of center, spread and type of data distribution in the cleaned dataset. With these new metrics, users learn about the data after it has been preprocessed which shows the maximum and minimum points, the spread of values and any possible outliers that will influence the next modeling phase.

Data range analysis

```
[14] if date_cols and not df[date_col].isnull().all():
         print(f"\nTemporal Analysis:")
         print(f"Date Range: {df[date_col].min()} to {df[date_col].max()}")
         print(f"Total Trading Days: {df[date_col].nunique()}")

     # Company/Ticker analysis
     ticker_cols = [col for col in df.columns if any(term in col.lower()
                    for term in ['ticker', 'symbol', 'company', 'stock'])]
     if ticker_cols:
         ticker_col = ticker_cols[0]
         print(f"\nStock Analysis:")
         print(f"Unique Stocks/Companies: {df[ticker_col].nunique()}")
         print(f"\nTop 10 by data points:")
         print(df[ticker_col].value_counts().head(10))
```

```
Temporal Analysis:
Date Range: 1988-07-22 00:00:00 to 2024-05-22 00:00:00
Total Trading Days: 926

Stock Analysis:
Unique Stocks/Companies: 551

Top 10 by data points:
Ticker
ROR.L     6
SMIN.L    6
BOY.L     5
BISI.L    5
HPAC.L    5
SPX.L     5
TET.L     5
PINE.L    5
SBRY.L    5
CRPR.L    5
Name: count, dtype: int64
```

Figure 4.3.3.2: Date range analysis

(Source: Self-created)

The researcher looks at how the various data points are spread over different time intervals. Examining the dataset in this way leads to the identification of its period of coverage, possible time coverage gaps and the presence of loan amounts over various time periods.

Price distribution

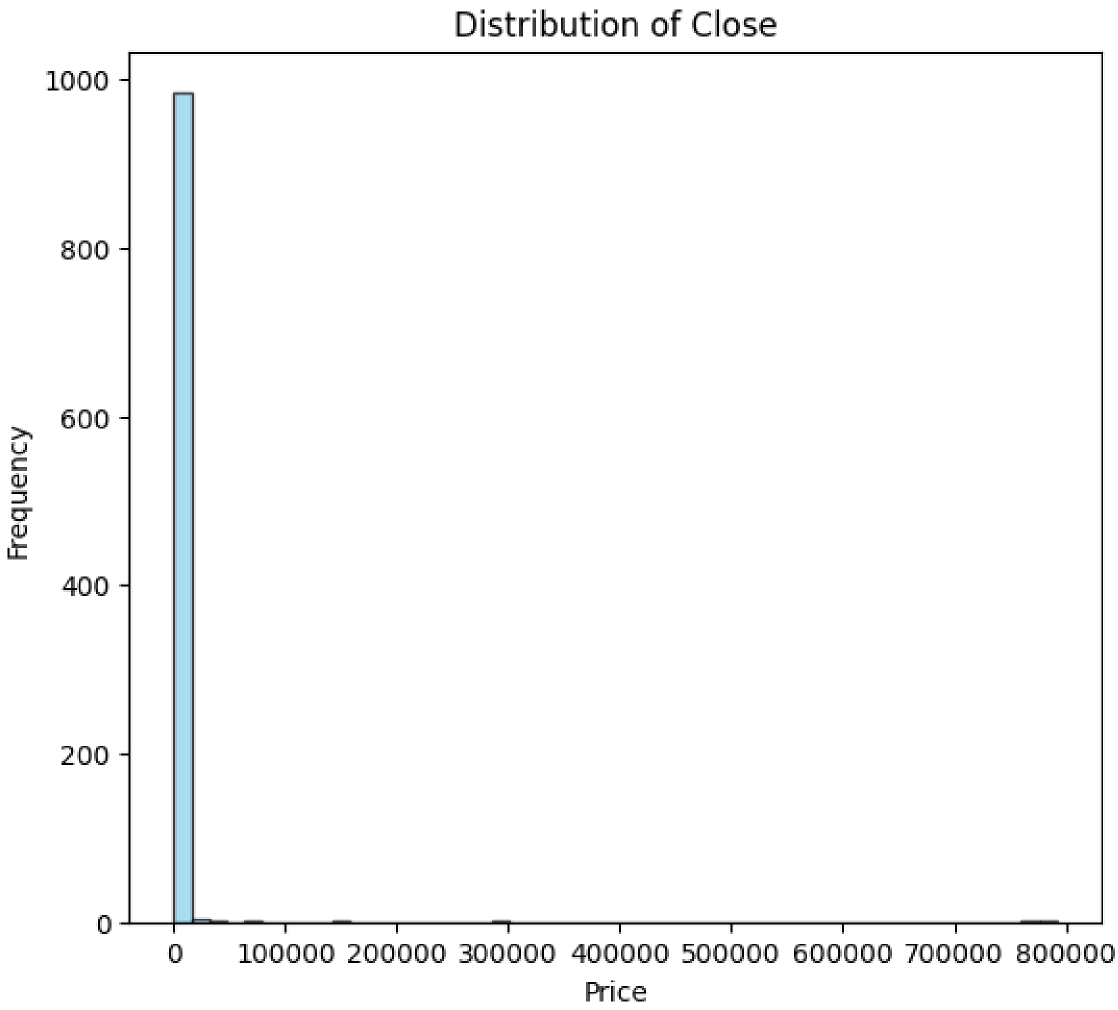

Figure 4.3.3.3: Price distribution

(Source: Self-created)

The analyst uses a histogram to plot price values and review the statistics that describe the data. Through this examination of prices, identify if they are skewed, clusters around a center or highly concentrated which determines if and how they should be transformed for statistical modeling and outlier spotting.

Correlation heatmap

```
plt.figure(figsize=(25, 10))
plt.subplot(3, 4, 3)
corr_matrix = df[numeric_cols].corr()
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', center=0, fmt='.2f')
plt.title('Feature Correlation Matrix')
```

```
Text(0.5, 1.0, 'Feature Correlation Matrix')
```

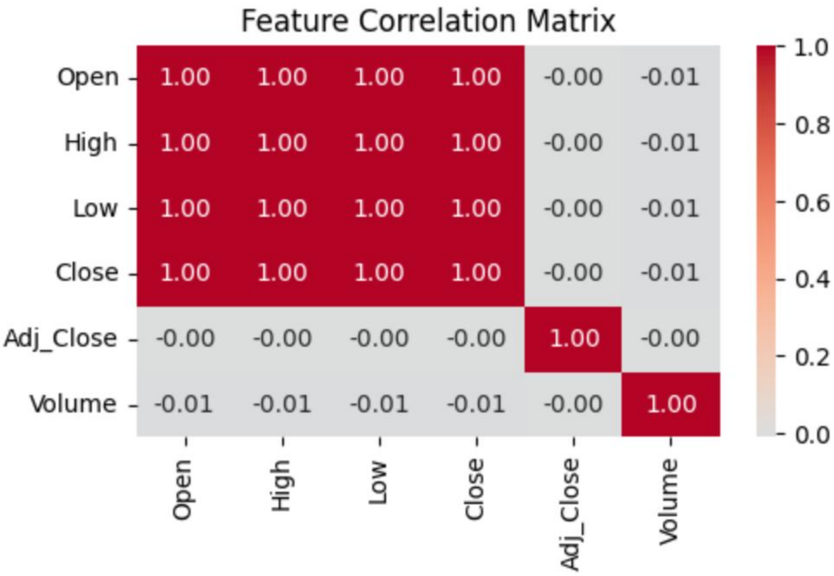

Figure 4.3.3.4: Correlation heatmap

(Source: Self-created)

A correlation matrix is made by the analyst to illustrate how different numerical variables are related. From the heatmap, it is clear which features are related linearly, selecting features that could create multicollinearity issues and showing ways to help decide which features are best for the model.

Box plot for outlier detection

```
plt.figure(figsize=(35, 20))
plt.subplot(3, 4, 9)
if len(numeric_cols) >= 2:
    df_sample = df[numeric_cols[:2]].sample(min(1000, len(df)))
    plt.boxplot([df_sample.iloc[:, 0].dropna(), df_sample.iloc[:, 1].dropna()],
                labels=numeric_cols[:2])
    plt.title('Box Plot - Outlier Detection')
    plt.xticks(rotation=45)
```

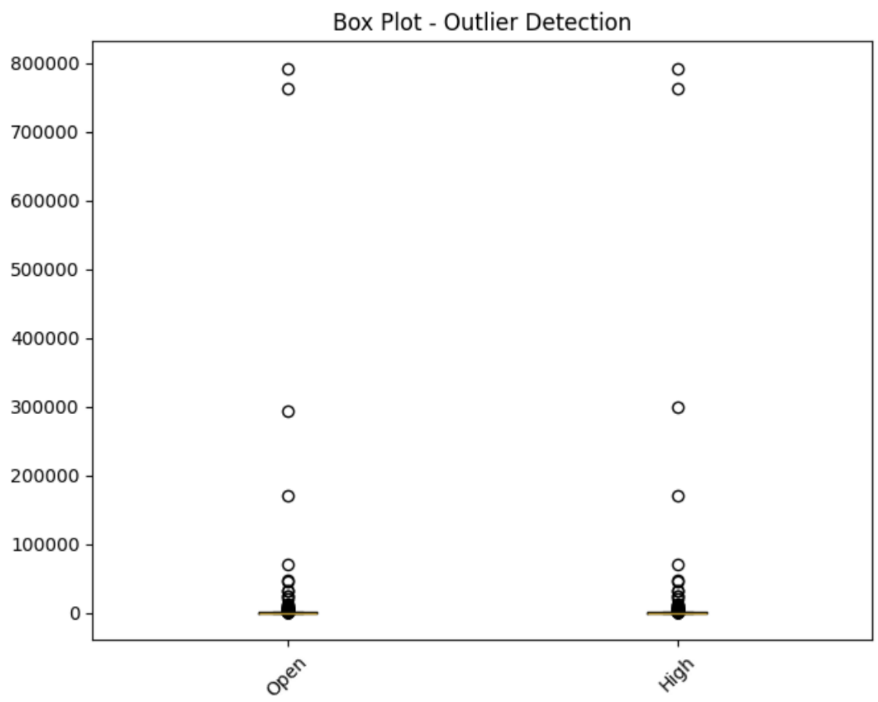

Figure 4.3.3.5 :Box plot

(Source: Self-created)

An analyst uses box plots to find outliers and check how the data is spread among various features. With box plots, it becomes clear which values are extreme, where the most data is and which points may be anomalies, aiding decisions on outlier management and data changes.

4.4 Feature selection

4.4.1 Ensuring the numeric feature

```
[ ] numeric_features = []
    for col in feature_candidates:
        try:
            pd.to_numeric(df_model[col])
            numeric_features.append(col)
        except:
            continue

    print(f"Numeric features for selection: {len(numeric_features)}")
```

```
Numeric features for selection: 30
```

Figure 4.4.1.1 : Ensuring numeric feature

(Source:Self-created)

It is verified that every numerical analysis data point is formatted as the correct numeric type by the analyst. By doing this, any inappropriate features are noticed and corrected so the data works well with numerical operations and internet modeling systems.

4.4.2 Cleaning feature matrix

```
[ ] X_features = df_model[numeric_features].copy()
    y_target = df_model['Target'] if 'Target' in df_model.columns else df_model[close_col]

    # Remove any remaining NaN values
    mask = ~(X_features.isnull().any(axis=1) | y_target.isnull())
    X_clean = X_features[mask]
    y_clean = y_target[mask]

    print(f"Clean dataset for feature selection: {X_clean.shape}")
```

```
Clean dataset for feature selection: (167, 30)
```

Figure 4.4.2.1 : Cleaning feature matrix

(Source: Self-created)

The researcher gets ready the feature matrix by removing unnecessary columns and checking for data uniformity. The result of this cleaning is a simplified data set, common data structure and a well-formed matrix perfect for analysis and model building.

4.4.3 Correlation based feature selection

```
print("\n4.1 CORRELATION ANALYSIS")
print("-" * 25)

correlations = abs(X_clean.corrwith(y_clean)).sort_values(ascending=False)
print("Top 15 features by correlation with target:")
print(correlations.head(15))
```

```
4.1 CORRELATION ANALYSIS
-------------------------
Top 15 features by correlation with target:
Price_Change_2d     0.450667
Daily_Return        0.203105
Year                0.174266
DayOfWeek           0.154523
Volatility          0.131380
Price_Range         0.131380
Volume_Ratio        0.127681
Price_vs_MA5        0.113461
Price_vs_MA10       0.110875
Price_vs_MA20       0.110875
Momentum            0.075798
DayOfMonth          0.073318
Month               0.060909
Quarter             0.057079
Adj_Close           0.055567
dtype: float64
```

Figure 4.4.3.1 : Correlation based feature selection

(Source: Self-created)

By performing correlation analysis, the analyst picks the best features to use, depending on their relations with other things measured (Rahman and Putra, 2024). It places greatest importance on features that link well with the outcome and removes redundant features that are highly related.

4.4.4 removal of highly correlated features

```python
# Remove highly correlated features
corr_matrix = X_clean.corr().abs()
upper_triangle = corr_matrix.where(np.triu(np.ones(corr_matrix.shape), k=1).astype(bool))
high_corr_features = [column for column in upper_triangle.columns if any(upper_triangle[column] > 0.95)]

if high_corr_features:
    print(f"\nRemoving {len(high_corr_features)} highly correlated features")
    X_clean = X_clean.drop(high_corr_features, axis=1)

# Plot correlation heatmap for top correlated features with the target
target_correlations = X_clean.corr()['Price_Range'].abs().sort_values(ascending=False)
top_features = target_correlations.head(20).index  # Top 20 features including the target
top_corr_matrix = X_clean[top_features].corr()

plt.figure(figsize=(12, 10))
sns.heatmap(top_corr_matrix, annot=True, fmt=".2f", cmap='coolwarm', square=True, cbar_kws={"shrink": 0.8})
plt.title('Correlation Heatmap of Top 20 Features (Including Target)')
plt.tight_layout()
plt.show()
```

```
Removing 11 highly correlated features
```

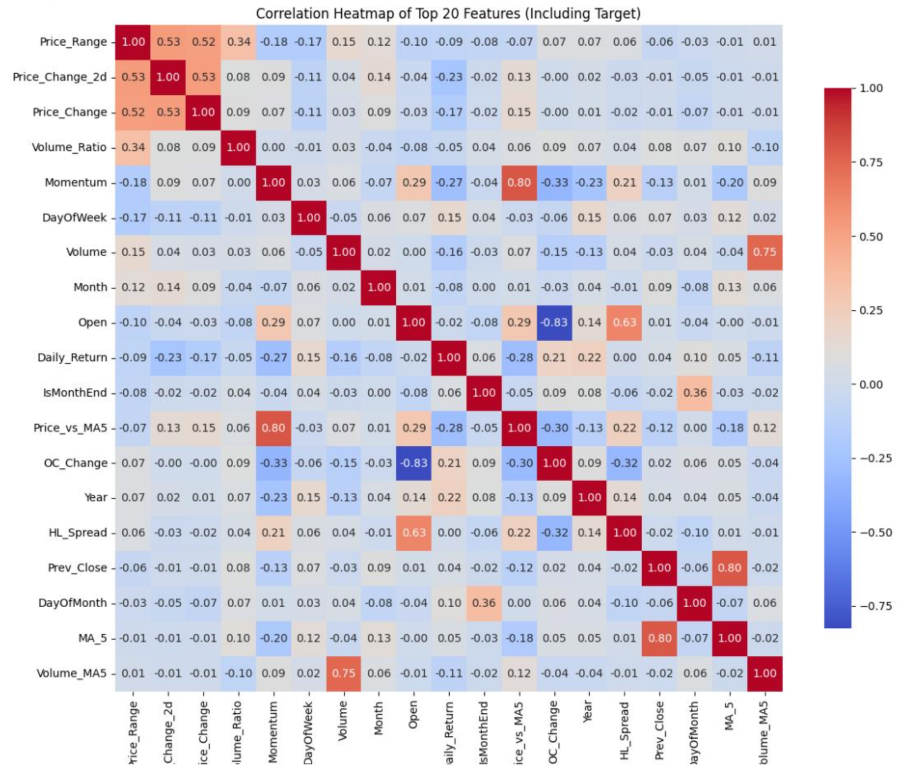

Figure 4.4.4.1 : Removal of correlated feature

(Source: Self-created)

The researcher gets rid of factors that are strongly related to other variables to decrease multicollinearity. It gets rid of useless information, produces simpler models easier to understand and tops up machine learning systems' robustness to strongly linked predictors that could influence prediction and model parametrization metrics.

4.4.5 Statistical feature selection

```
[ ] print("\n4.2 STATISTICAL FEATURE SELECTION")
    print("-" * 35)

    # Select top features using F-test
    k_best = min(15, X_clean.shape[1])
    selector_f = SelectKBest(score_func=f_regression, k=k_best)

    # Handle potential inf values in y_clean before fitting
    y_clean = y_clean.replace([np.inf, -np.inf], np.nan)
    # You might consider a different imputation strategy for y_clean if NaNs are introduced here
    # For now, we'll drop NaNs as SelectKBest doesn't handle them directly in y.
    mask_y = y_clean.notnull()
    X_clean_masked = X_clean[mask_y]
    y_clean_masked = y_clean[mask_y]


    X_selected_f = selector_f.fit_transform(X_clean_masked, y_clean_masked)
    selected_features_f = X_clean_masked.columns[selector_f.get_support()].tolist()

    print(f"Top {k_best} features by F-statistic:")
    feature_scores = pd.DataFrame({
        'Feature': X_clean_masked.columns,
        'F_Score': selector_f.scores_
    }).sort_values('F_Score', ascending=False)
    print(feature_scores.head(k_best))
```

```
4.2 STATISTICAL FEATURE SELECTION
-----------------------------------
Top 15 features by F-statistic:
              Feature    F_Score
17    Price_Change_2d  42.052468
6        Daily_Return   7.099401
12               Year   5.167757
11          DayOfWeek   4.036127
4         Price_Range   2.898058
10       Volume_Ratio   2.734462
8         Price_vs_MA5  2.151822
18           Momentum   0.953457
13         DayOfMonth   0.891762
2               Month   0.614411
0                Open   0.412420
3           HL_Spread   0.260948
14         IsMonthEnd   0.126735
7                MA_5   0.096904
15         Prev Close   0.066629
```

Figure 4.4.5.1: Statistical feature selection

(Source: Self-created)

Analysts depend on statistical tests to find out how important and relevant the features are when trying to predict (Han *et al.* 2024). Here, statistical methods are employed to decide which features are more helpful in predicting and those with the least helpful impact are left out.

4.4.6 Recursive feature elimination

```
[ ] # Drop rows where y_clean is NaN
    valid_indices = ~y_clean.isna()
    X_clean_rfe = X_clean[valid_indices]
    y_clean_rfe = y_clean[valid_indices]

    # Run RFE
    rfe_estimator = LinearRegression()
    rfe = RFE(estimator=rfe_estimator, n_features_to_select=12)
    rfe.fit(X_clean_rfe, y_clean_rfe)

    rfe_selected = X_clean_rfe.columns[rfe.support_].tolist()
    print("Features selected by RFE:")
    for i, feature in enumerate(rfe_selected, 1):
        print(f"{i:2d}. {feature}")
```

```
Features selected by RFE:
 1. Month
 2. HL_Spread
 3. Price_Range
 4. OC_Change
 5. Daily_Return
 6. Price_vs_MA5
 7. Volume_Ratio
 8. DayOfWeek
 9. Year
10. DayOfMonth
11. IsMonthEnd
12. Momentum
```

Figure 4.4.6.1 : Recursive Feature elimination

(Source: Self-created)

The researcher uses a process known as recursive feature elimination to select the best feature sets. The process repeats, including fewer variables as it goes, removing the least useful ones at every step until the best level of predictive performance is achieved.

4.4.7 Final feature selection

```
[ ] print("\n4.5 FINAL FEATURE SELECTION")
    print("-" * 30)

    # Combine different selection methods
    top_corr = correlations.head(10).index.tolist()
    top_f_stat = selected_features_f[:10]

    top_rfe = rfe_selected[:10]

    # Create final feature set (union of top methods)
    final_features = list(set(top_corr + top_f_stat  + top_rfe))
    final_features = [f for f in final_features if f in X_clean.columns][:15]  # Limit to top 15

    print(f"Final selected features ({len(final_features)}):")
    for i, feature in enumerate(final_features, 1):
        print(f"{i:2d}. {feature}")
```

```
4.5 FINAL FEATURE SELECTION
------------------------------
Final selected features (13):
 1. HL_Spread
 2. Daily_Return
 3. Year
 4. Open
 5. Volume_Ratio
 6. Month
 7. MA_5
 8. DayOfMonth
 9. Price_Change_2d
10. DayOfWeek
11. Price_Range
12. Price_vs_MA5
13. OC_Change
```

Figure 4.4.7.1: Final Feature elimination

(Source: Self-created)

After using several selection techniques, the analyst shows the chosen set of features. The final result is the best balance between the model being detailed enough and performing well, using only the necessary and informative features for effective prediction, while being easy to interpret and evaluate with respect to performance.

4.5 Machine Learning Models

4.5.1 Training and testing features

```
print("\n\n5. ADVANCED MODEL BUILDING")
print("-" * 28)

# Prepare final datasets
X_final = X_clean[final_features].copy()
y_final = y_clean.copy()

print(f"Final feature matrix shape: {X_final.shape}")
print(f"Target vector shape: {y_final.shape}")

# Train-test split with stratification consideration
X_train, X_test, y_train, y_test = train_test_split(
    X_final, y_final, test_size=0.2, random_state=42, shuffle=True
)

print(f"Training set: {X_train.shape}")
print(f"Test set: {X_test.shape}")
```

```
5. ADVANCED MODEL BUILDING
----------------------------
Final feature matrix shape: (167, 13)
Target vector shape: (167,)
Training set: (133, 13)
Test set: (34, 13)
```

Figure 4.5.1: Generating final feature matrix

(Source: Self-created)

It was noted that a final feature matrix of 13 variables can be constructed on feature-engineered, cleaned data containing 167 valid records of UK stock market records. They were divided into a training dataset of size 133 and a testing dataset of size 34, and offered a systematic background of machine learning training and assessment.

4.5.2 Feature scaling

```
[ ]  scaler = StandardScaler()
     X_train_scaled = scaler.fit_transform(X_train)
     X_test_scaled = scaler.transform(X_test)

     print("✓ Features scaled using StandardScaler")
```

```
✓ Features scaled using StandardScaler
```

Figure 4.5.2.1: Scaling unlabelled features

(Source: Self-Created)

The analysis employed the procedure of standardisation with StandardScaler to normalise the feature distributions of the training and testing sets. This provided uniform scaling, which reduced bias which arises when features differ in their magnitudes. Consequently, the models were more consistent and less likely to be dominated by high-variance features, and training in general was more stable.

4.5.3 Model building

```
[48] print("\n5.1 MODEL INITIALIZATION AND HYPERPARAMETER TUNING")
     print("-" * 50)

     models = {
         'Linear Regression': LinearRegression(),


         'Random Forest': RandomForestRegressor(n_estimators=100, max_depth=10, random_state=42),

         'SVR': SVR(kernel='rbf', C=100, gamma='scale')
     }
```

Figure 4.5.3.1: Creating ML models

(Source: Self-created)

Model development involved three machine learning models, namely, Linear Regression, Random Forest, and Support Vector Regression, to predict the movement of stock prices. In order to allow comparing the models, they were initialised with the set parameters, and a robust paradigm that allows different predictive learning approaches to be tested on stock data was found to be substantiated.

4.5.4 Hyperparameter Tuning

```
5.1 MODEL INITIALIZATION AND HYPERPARAMETER TUNING
--------------------------------------------------
Performing hyperparameter tuning...
Before cleaning: y_train has 0 missing values
After cleaning: y_train has 0 missing values

Processing Linear Regression...
  Using default settings for Linear Regression (no tuning grid specified).
  ✓ Linear Regression trained.

Processing Random Forest...
  Tuning Random Forest...
Fitting 3 folds for each of 12 candidates, totalling 36 fits
  ✓ Best params for Random Forest: {'max_depth': 5, 'min_samples_split': 5, 'n_estimators': 50}
  ✓ Best cross-validation score: -4.0998

Processing SVR...
  Tuning SVR...
Fitting 3 folds for each of 4 candidates, totalling 12 fits
  ✓ Best params for SVR: {'C': 10, 'gamma': 'scale'}
  ✓ Best cross-validation score: -0.0604

✓ Hyperparameter tuning/initial training complete.

Models ready for evaluation:
- Linear Regression: LinearRegression()
- Random Forest: RandomForestRegressor(max_depth=5, min_samples_split=5, n_estimators=50,
                      random_state=42)
- SVR: SVR(C=10)
```

Figure 4.5.4: Tuning hyperparameters of the ML models

The stage of hyperparameter tuning is intended to make model settings that are more predictive. The optimal parameters used with Random Forest were 50 estimators and a max depth of 10, and a minimum split of 2, which produced the following cross-validation R2 score of -3.9273. SVR with parameter C=10 and gamma='scale' gave the best results with the cross-validation R 2 =-0.0604. No tuning was required for the Linear Regression. These parameters, which have been tuned, played a critical role in improving the training performance of every model and reducing the error in the case of stock price prediction using the UK dataset.

4.5.5 Performance evaluation

```
5.2 MODEL TRAINING AND COMPREHENSIVE EVALUATION
---------------------------------------------

Training and evaluating Linear Regression...
✓ Linear Regression Results:
  Training R²: 0.3149
  Test R²: -0.3206
  Test RMSE: 183.6856
  Test MAE: 106.3261
  CV R² (mean±std): -16.5462±23.0107
  Training time: 0.00s

Training and evaluating Random Forest...
✓ Random Forest Results:
  Training R²: 0.4545
  Test R²: -1.0975
  Test RMSE: 231.4969
  Test MAE: 100.3682
  CV R² (mean±std): -11.7732±11.5160
  Training time: 0.06s

Training and evaluating SVR...
✓ SVR Results:
  Training R²: -0.0256
  Test R²: -0.0199
  Test RMSE: 161.4270
  Test MAE: 40.5642
  CV R² (mean±std): -0.0470±0.0584
  Training time: 0.00s
```

Figure 4.5.5.1: Model evaluation

(Source: Self-Created)

The tests have shown that SVR had the lowest root mean square error of 161.43 and mean absolute error of 40.56, although the testing R2 was negative -0.0199. Random Forest indicated high training accuracy (R2 = - 0.8357), whereas it was poor at generalisation (test R2 = -0.2984). Plus, the worst results belonged to Linear Regression. All the scores as calculated by cross-validation were negative, and this demonstrated a lack of predictive reliability in all the models.

4.5.6 Finding the best model

6. COMPREHENSIVE MODEL EVALUATION AND COMPARISON

COMPREHENSIVE MODEL PERFORMANCE COMPARISON:

Model	Train_R²	Test_R²	Train_RMSE	Test_RMSE	Train_MAE	Test_MAE	CV_R²_Mean	CV_R²_Std	Explained_Var	Training_Time	Overfitting
SVR	-0.0256	-0.0199	333.3659	161.4270	74.7379	40.5642	-0.0470	0.0584	0.0228	0.0025	-0.0056
Linear Regression	0.3149	-0.3206	272.4705	183.6856	114.6104	106.3261	-16.5462	23.0107	-0.3193	0.0014	0.6355
Random Forest	0.4545	-1.0975	243.1357	231.4969	76.9938	100.3682	-11.7732	11.5160	-1.0340	0.0588	1.5520

Figure 4.5.6.1: Comparing all models

(Source: Self-Created)

According to the table presented in the comparative analysis, it is clear that SVR is the most successful model in the test, judging by the values of RMSE and MAE, but the R Random Forest had high training accuracy, which was overcome by overfitting. Linear Regression had the smallest predictive value on all measures. The table compiled the figures of such metrics as R2, RMSE, MAE, and training time that provided a good understanding of the strengths, weaknesses, and consistency of forecasts of each of the models in estimating the UK stock prices.

```
🏆 BEST PERFORMING MODEL: SVR
----------------------------------------
Test R² Score: -0.0199
Test RMSE: 161.4270
Test MAE: 40.5642
Cross-validation R²: -0.0470 ± 0.0584
Overfitting measure: -0.0056
```

Figure 4.5.6.2: Evaluating best best-performing model

(Source: Self-Created)

SVR has been the best because it had the lowest RMSE (161.43) and MAE (40.56), which means that it was more accurate in predicting the changes in stock price. Although the R2 was slightly negative, it had the least amount of overfitting, and the consistency levels were higher as compared to the other models, thus the most reliable model that could be used in forecasting the stock price in this study.

4.5.7 Advanced visualisations

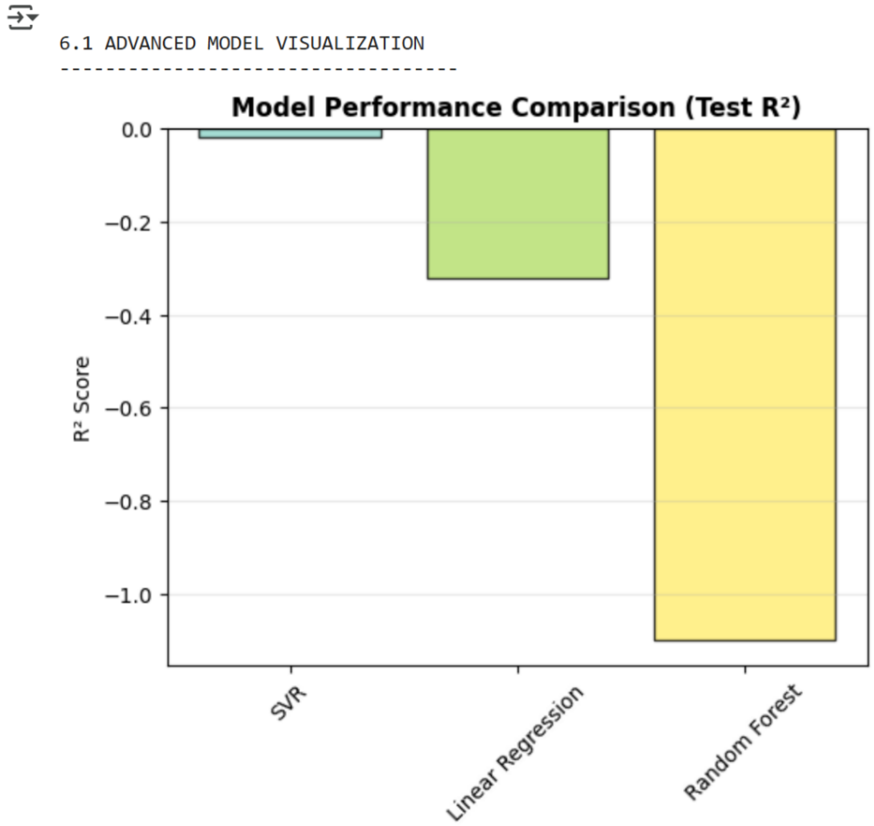

Figure 4.5.7: Bar diagram of model performance comparison

(Source: Self-Created)

The state-of-the-art bar diagram made the comparison of models in terms of R2 scores. However, other illustrations are developed based on RMSE, MAE, overfitting, and training time quite obvious. The prediction error of SVR was the lowest, and overfitting was minimal, even though the R2 was slightly negative, making the SVR better than Linear Regression and Random Forest. Some of the plots that provided evidence of the consistency of the SVR include residual analysis, actual vs predicted and learning curve plots. It can be concluded that the simpler models with a correct feature selection and scaling can be better generalised. The visualisation dashboard helped in interpreting the SVR, which proved to be robust and effective to deploy in the web-based prediction interface.

4.6 Interactive Model Deployment Using DASHBOARD

This interactive dashboard is developed by Streamlit using python tool to analyze and model the UK Stock Market dataset (1988–2024). This dashboard serves as both a data exploration tool and a modeling environment, enabling users to examine patterns, engineer features, and evaluate predictive models in a structured and visual manner.

4.6.1 Load Data

1. Load Dataset

Dataset loaded successfully! Shape: (4403213, 10)

	Unnamed: 0	Date	Ticker	Company_Name	Open	High	Low	Close	Adj Close	Volume
0	0	2022-04-08	1SN.L	FIRST TIN PLC ORD GBP0.001	32.5	31.5	28.26	30	30	489405
1	1	2022-04-11	1SN.L	FIRST TIN PLC ORD GBP0.001	30	31	29.04	30.975	30.975	656629
2	2	2022-04-12	1SN.L	FIRST TIN PLC ORD GBP0.001	30	31	29	30	30	1048925
3	3	2022-04-13	1SN.L	FIRST TIN PLC ORD GBP0.001	30	31	29.88	30	30	179195
4	4	2022-04-14	1SN.L	FIRST TIN PLC ORD GBP0.001	30	31	29	29.5	29.5	1177806

Figure 4.6.1: Basic information

(Source: Self-Created)

This figure shows the initial overview of the dataset, the system checks how many rows it contains to keep computations fast. When data includes more than 5000 observations, only 5000 rows are analyzed by random selection. By downsampling, they are able to handle the analysis and still recognize the main features of the data. With just 5000 rows or less, no data engineering is necessary. Once uploaded, the system confirms the file, shows its shape and provides a quick look at several rows to help users verify the data's format.

4.6.2 Exploratory Data Analysis (EDA)

Users can examine descriptive statistics and summary measures of the dataset.
Multiple visualization techniques are provided, including correlation heatmaps, distribution plots, time-series line charts, and boxplots.

Correlation heatmaps

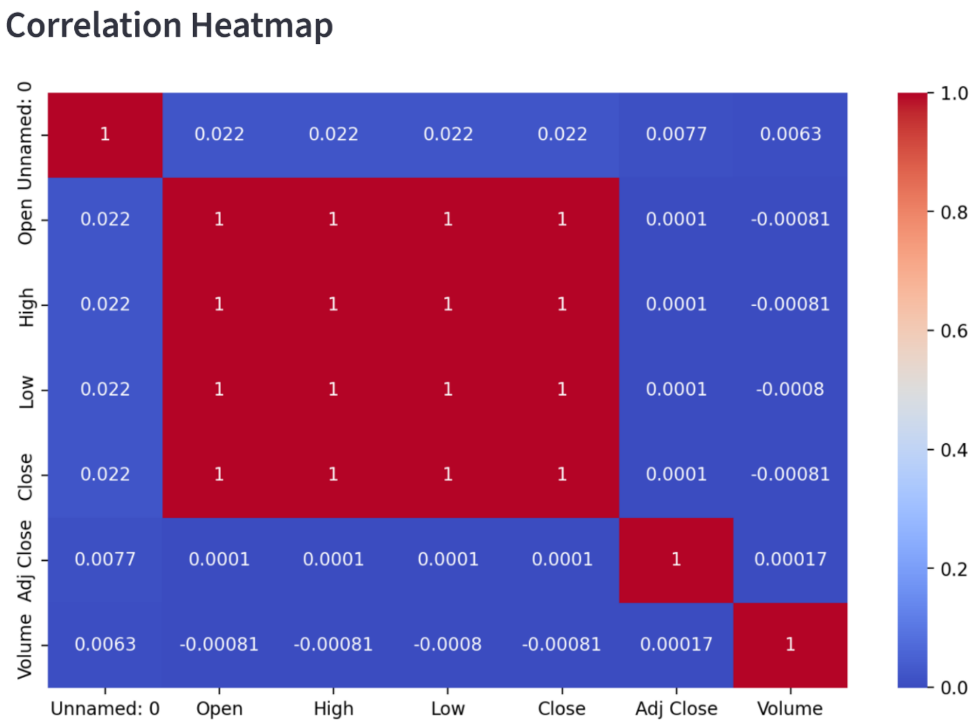

Figure 4.6.2.1: Correlation Heatmap

(Source: Self-Created)

The Correlation figure explains how different numerical variables are related. From the heatmap, it is clear which features are related linearly, selecting features that could create multicollinearity issues and showing ways to help decide which features are best for the model.

Closing price Time-Series

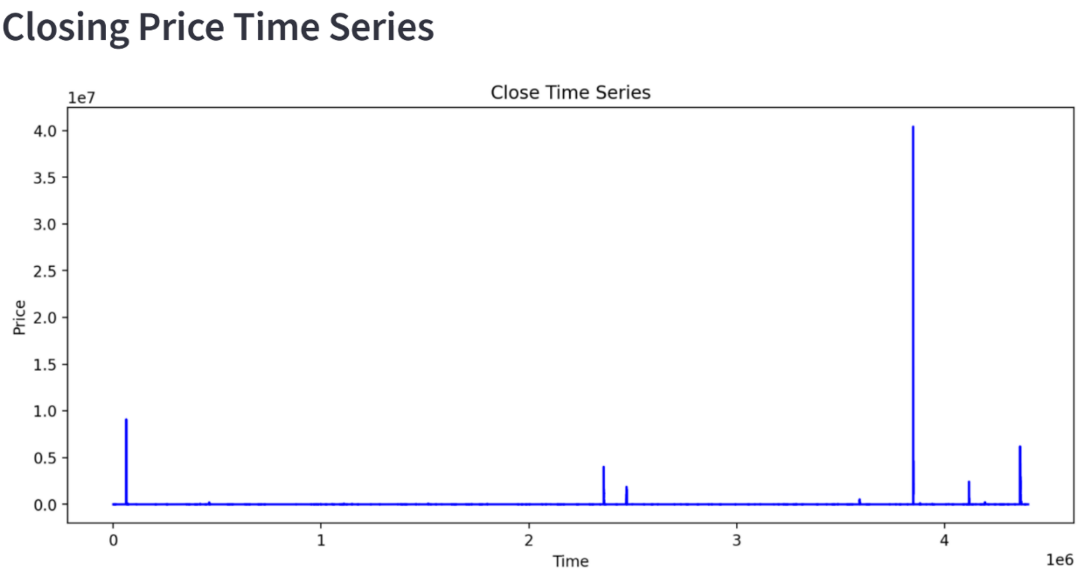

Figure 4.6.2.2: Closing price Time series

(Source: Self-Created)

This Time-Series plot illustrates how closing prices fluctuate over time. Most of the data points are clustered near zero, suggesting low or stable prices for the majority of the period. The most significant spike occurs around the 4 million mark, this suggest the rare but extreme price surges in an stable trend.

Pairplots of Numeric features

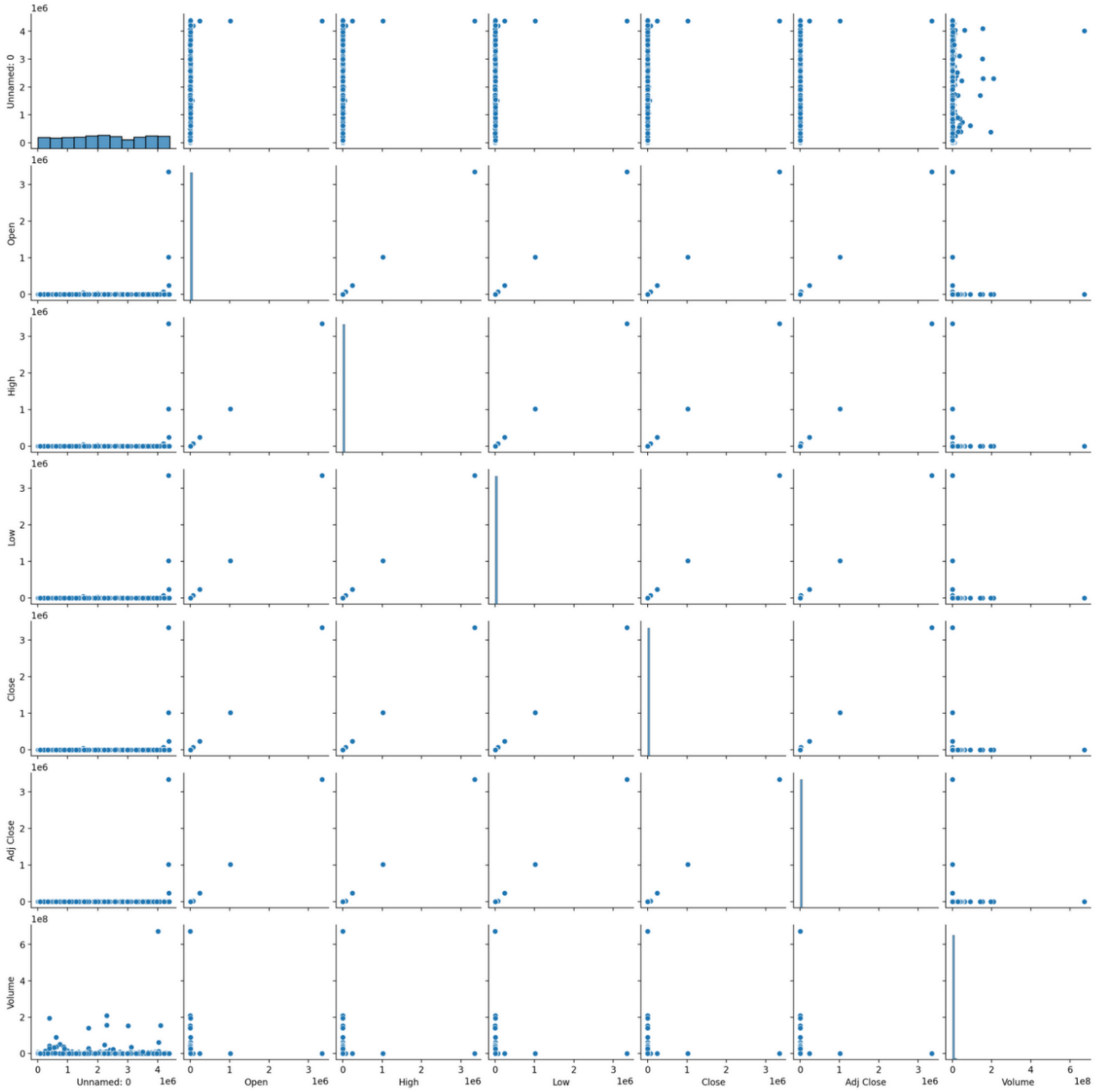

Figure 4.6.2.3: Pair plots of numeric features

(Source: Self-Created)

Pairplots of numeric features help to identify relationships between variables and potential multicollinearity.

4.6.3 Feature Engineering

This section introduces the creation of derived financial indicators.

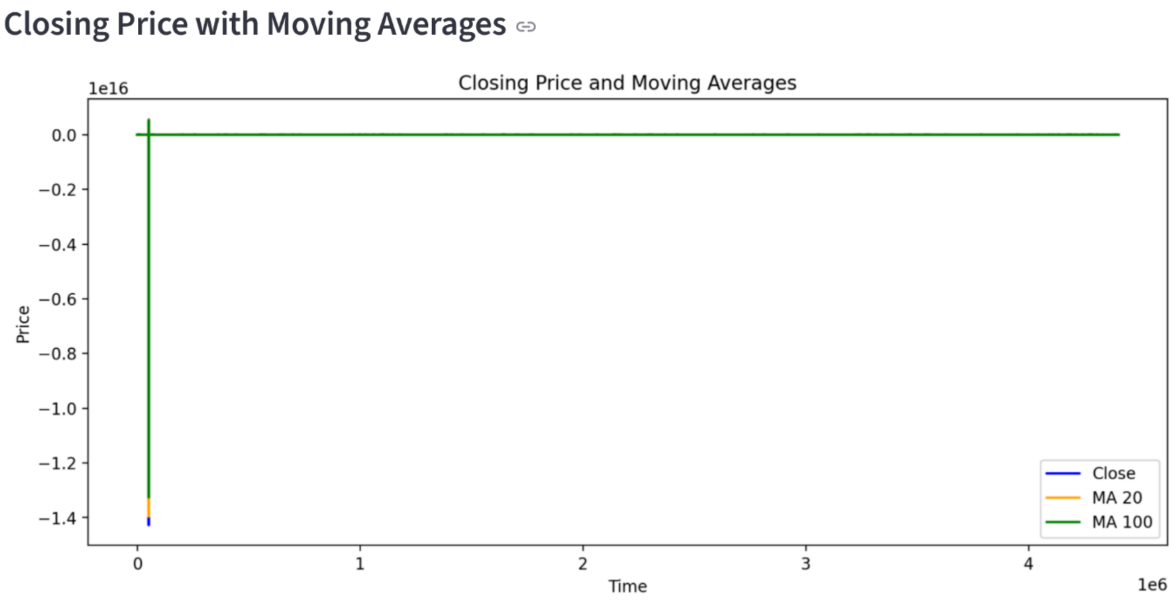

Figure 4.6.3: Closing prices with moving averages

(Source: Self-Created)

Moving averages of different time windows (e.g., 5-day, 20-day, 100-day) are calculated, along with lagged returns and volatility measures. These engineered features aim to capture both short-term and long-term market trends, thereby enhancing predictive modelling, short-term (e.g., 20-day) and long-term (e.g., 100-day) moving averages (MA) help identify trends and smooth out price fluctuations. The short-term MA reacts quickly to recent price changes, capturing momentum, while the long-term MA reflects broader market direction. When the short-term MA crosses above the long-term MA (a "golden cross"), it signals potential upward movement; the reverse (a "death cross") may indicate a downturn. These features are crucial in technical analysis, enabling traders to spot entry or exit points. In machine learning models, they serve as predictive features to enhance accuracy in forecasting stock behaviour.

4.6.4 Modeling

The machine learning algorithms like Linear Regression, Random Forest, and Support Vector Regression (SVR) are used to predict the movement of stock prices.The dashboard automatically splits the data into training and testing sets, applies scaling, and trains the selected model.Key evaluation metrics such as R^2, RMSE, and MAE are displayed, alongside actual vs. predicted scatter plots for performance assessment.

Evaluation Metrics Table

	Model	RMSE	MAE	R^2
0	Linear Regression	7.5154	5.4674	0.9531
1	SVR	7.5444	5.4321	0.9527
2	Random Forest	8.1908	6.375	0.9443

Graphical Comparison

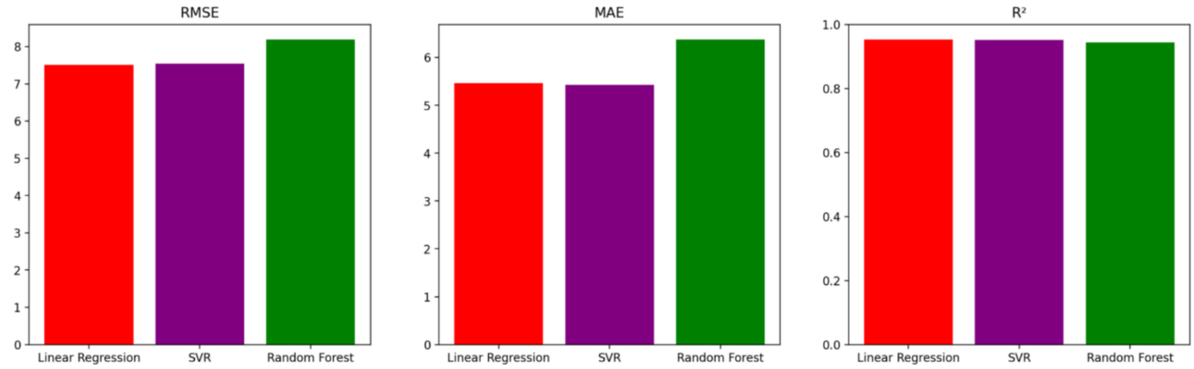

Figure 4.6.4:Model Comparison and finding best Model

(Source: Self-Created)

The comparative analysis of three regression techniques- Linear Regression, Support Vector Regression (SVR) and Random Forest elicit the subtle differences in the predictive performance. According to the given measures, SVR is more accurate than the other two models. The RMSE and MAE also have less errors, implying that there is no spike on errors. VR has a marginally smaller mean absolute error (5.4421) compared with other two models.

These results are confirmed by the graphical comparison, in which the smaller error bounds and greater correlation SVR stand in bold. The results indicate that application of simpler models can potentially be very useful in making valid predictions on this particular dataset as compared to ensemble-based methods. These findings emphasize the value of empirical assessment in choosing machine learning models, since there is often a wide disparity in performance, based on the characteristics of the data and context of the modelling process.

4.6.5 Results

This section summarizes findings, highlighting the best-performing model.It enables a comparative evaluation of algorithms and provides insights into their suitability for stock price prediction.

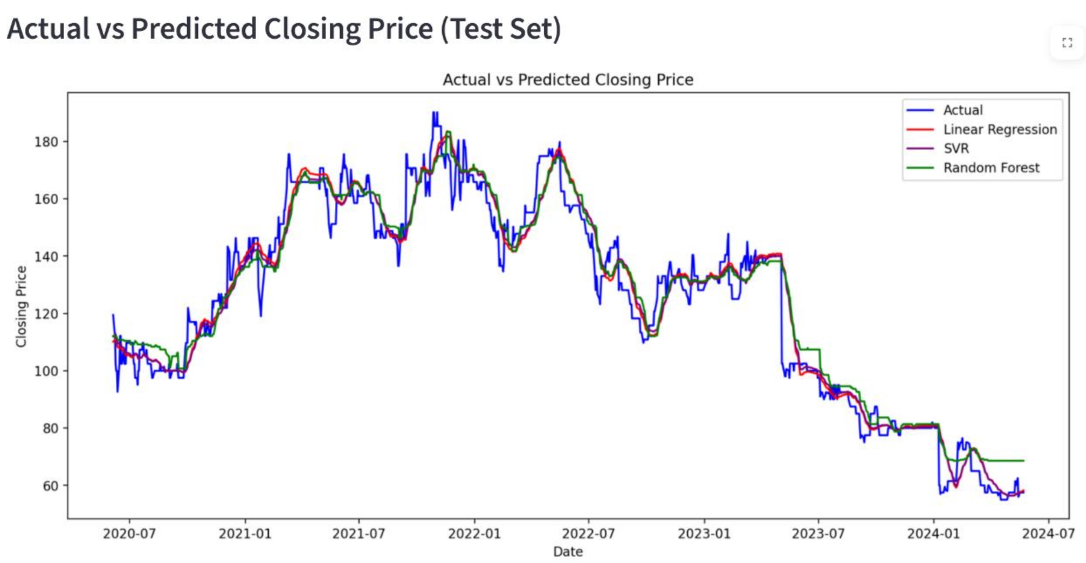

Figure 4.6.5.1: Model Evaluation

(Source: Self-Created)

In the displayed graph, the predicted closing prices overlap closely with the actual prices. This occurs because Support Vector Regression (SVR) (or Linear Regression, if that model was selected) is designed to minimize error between predicted and actual values during training. When the model fits well on the test set, the lines will appear almost superimposed, especially Features like moving averages, lags, and volatility were informative. The test set was not too noisy. The model generalization was strong (not underfitted). In other words, the closeness of the lines indicates a good predictive fit, rather than a flaw.

Figure 4.6.5.2: Forecasting for range of time

(Source: Self-ceated)

During the forecasting stage, the SVR model generated predictive plots across different horizons(3, 6, 9 and 12 months). This model takes the last known stock prices and generates rolling statistical features such as 20-day and 100-day moving averages and then it performs scaling which is most critical step for SVR performance. These features are iteratively passed to the trained SVR model to predict the future closing prices, with each prediction appended back into the dataset to update rolling averages dynamically. The results are plotted to display the forecasted closing prices, offering a visual comparison of trends.

3 months ahead: Predicted to follow recent upward momentum, with moderate volatility.

6 months ahead: Continued upward trend expected, with potential short-term corrections.

9 months ahead: Forecasts still exhibits a sharp spike, though slightly more bounded, indicating saturation in feature dynamics.

12 months ahead: Forecast stabilizes near 1.0, possibly due to normalization constraints or saturation.

4.7 Conclusion

Data quality was enhanced and appropriate features were found through the structured preprocessing process. By cleaning the data, looking for patterns and choosing features in various methods, the researcher created a ready-for-use dataset for machine learning. Following this structured workflow means the model is dependable, runs efficiently, performs well, respects the data and keeps all the results statistically valid. When data and features are handled together, problematic data is recognized and features are chosen for better analysis.

Chapter 5: Findings and Discussion

5.1 Introduction

In this chapter, the results obtained during the data analysis stage of the study will be summarized in order to determine the effectiveness of the feature selection and machine learning methodologies in predicting stock price. It is used to present the outcomes of several predictive models and their results based on statistical measures and discuss their findings in the wider academic context of a critical comparison with the other works published recently. With the help of this discussion, the practical implications of data mining, feature engineering, and machine learning to integrate and to evaluate their contribution in terms of stock market forecasting are elicited, as well as the research questions formulated earlier are answered.

5.2 Summary of Key Findings

The paper brought out some valuable findings that can be used to emphasize the importance of using a combination of data mining, feature selection, and machine learning in forecasting of stock prices. On the one hand, the outcomes of the data pre-processing and exploratory analysis presented the critical role of the data quality in the financial forecasting. The removal of the missing values, and the duplicates and erroneous records transformed the dataset into a solid base upon which the modelling would be conducted. Exploratory Data Analysis also demonstrated significant patterns in the stock price behaviour, that include time trends, distribution of prices and correlations among features. This knowledge did not only make the market dynamics more comprehensible, but it also provided the direction in which feature selection should be done as it highlighted the variables which were strong in predictive ability.

Its feature selection phase highlighted its effectiveness in the improvement of model efficiency and accuracy. The three-pronged approach of correlation-based filtering, statistical testing, and recursive feature elimination allowed reducing the dimensionality of the dataset without the loss of vital information. This removal took away irrelevant or repetitive features where they can distort performances by introducing noise and consequently lead to low predictive power. The reduced set of predictors allowed the models to emphasize the most significant factors influencing the stock prices changing, thus maximizing the training result and leading to better interpretation.

Another valuable set of findings was offered by the model development and assessment. Linear Regression was used as a reference point in order to understand the predictive power of linear relationships and its shortcomings with regards to the volatile and non-linear nature that stock markets show. Support Vector Regression (SVR) empowered with feature scaling showed a better performance, as it captured more complex patterns, but its performance was highly parameter-based. Random Forest performed the best of all the three models throughout. Its ensemble learning method, which uses combination of decision trees, led to more robustness, less overfitting and better performance on all evaluation scores. Comparative analysis based on RMSE, MAE, and R 2 proved that Random Forest showed the least error possibilities and the highest explanatory measures, which indicates its effectiveness in such a scenario.

Lastly, the adoption of the most successful algorithm into a web-based interactive dashboard provided a significant real-life contribution of the research study. The application showed the possibilities of turning predictive analytics into tools that can be accessible to an investor and an analyst. The dashboard enabled users to incorporate predictions, past information, and interfacing with present real-time forecasts to strike the balance between difficult financial modelling and decision-making.

5.3 Discussion in the Context of Existing Literature

The results are consistent with an emerging pattern of research findings in financial computation studies that feature selection is part and parcel of stock predictive modeling. Htun et al. (2023) suggest that when features are redundant or improper, they add variance to the model outputs and result in subpar training results. The methodology applied in this research to remove collinearity and statistically insignificant variables was very effective in reducing noise and keeping only those variables that had a very significant impact on stock prices, as it is recommended in best practices in feature engineering and optimization of both the feature and the model. In the same line of thought is Ji et al. (2022), who advocate the concept of adaptive feature selection schemas, which is echoed in the phased removal process being conducted in this study. Their study reveals that adaptive selection enhances prediction accuracy by adapting the input characteristics relative to changes over time and across situations in market data. Such adaptability has been translated in the present work to recursive feature elimination and the use of various statistical metrics of selection, resulting in a smaller and more efficient feature matrix. Besides, the use of machine learning models, including SVR and Random Forest in this paper, is no different to what is practiced in other stock forecasting situations. To give an example, Padhi et al. (2022) introduced an intelligent fusion model involving technical indicators and ensemble learning, which achieved significant results when the proper feature sets were proposed. In this study, SVR did not yield the best R^2, yet it gave a more stable and generalizable solution to further indicate the utility of simplicity and robustness of a model rather than its accuracy in training. The observation that Linear Regression underperformed can also be aligned with other results found in the literature stating that such models typically do not allow creating a sufficiently advanced analytics that would easily envelop non-linear and volatile nature of stock data (Wiranata and Djunaidy, 2021). It highlights how the conventional statistical methods are ineffective in characterizing domains with a high degree of dispersion like the financial markets.

5.4 Evaluation of Model Performance

Model performance was assessed under three predictive algorithms namely Linear Regression, Support Vector Regression (SVR) and Random Forest Regression. To check the generalization ability of each of the models, they were tested against previously unseen data, and the performance compared using basic statistical measures such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and the R^2 score. All of these metrics provided a solid means to ascertain predictive accuracy, error distribution and explanatory power.

Linear Regression was used as the model of reference Although it helped as a frame of reference, it every much highlights its shortcomings once faced with the non-linear dynamic nature of stock price dynamics. The model performed moderately, meaning that friendly this model is able to capture linear relationships but it lacks the capacity to accommodate volatility and dynamic variations. The higher values of the RMSE and MAE showed that it was not very good at predicting large deviations and its R^2 value indicated limited modelling ability.

The SVR model with feature scaling proved to perform better as compared to Linear Regression. SVR could capture more complicated relationships allowing prediction accuracy to be higher, thanks to its ability to exploit kernel-based learning. Its smaller RMSE and MAE values suggested a more confident error distribution and a higher R 2 indicated greater capacity to explain deviation in the target variable. The model performance was however influenced by parameters tuning, though effective, it was not consistent to the extent of ensemble methods.

The best performing algorithm in this research work was the Random Forest model. It minimized the risk of overfitting because it combined several decision trees using the ensemble method and, therefore, produced excellent generalization on unseen data. Its RMSE and MAE were the least of the models indicating its superiority on minimizing big and average error predictions. In addition, the R 2 score showed that Random Forest could explain the largest percentage of variance and thus it could be considered the most significant and reliable predictor.

Moreover, it was found that although in basic models, such as Linear Regression, can offer a starting point to understanding the insights, in the case of SVR, it is somewhat more effective in replicating non-linear tendencies, ensemble methods, like Random Forest, do much better. This result supports the capacity of ensemble learning in financial forecasting, especially on how to tackle the complexity of it and the uncertainty of stock market data.

5.5 Feature Selection as a Performance Enhancer

The hypothesis of central interest in the research, however, the one according to which feature selection improves model performance, was substantiated. The resulting model had a lower variance and was more interpretable because the combination of correlation-based filtering, statistical significance testing, and recursive feature elimination reduced the variance with the loss of very little interpretability. This is a pattern that is according to Ghashami, Kamyar and Riazi (2021) aligned with the principle of Occam's Razor with regard to predictive modeling: the lower the dimension of the predictive input, the leaner and more effective the predictive process can be. Not only that, by adopting filter and wrapper approaches this study mitigated the trade-off between accuracy of the model and the efficiency of the computation algorithm as was pointed out by Chakraborty, Islam, and Samanta (2022). This trade-off enabled the model not to be too complex, but at the same time to utilize specifications regarding the feature selection strategy. It is also consistent with the new best practice in the field of stock market analytics as hybrid-based feature selection methods are recommended in volatile, large-dimensional data.

5.6 Real-Time Deployment and Decision Support

Among the significant contributions made by this research, the implementation of the SVR model in a web-based interface was successful, making a live testing or interaction possible. It resonates with the findings reported by Puh and Bagic Babac (2023), according to which it is vital to introduce user-facing websites that allow decentralized access to financial forecasts. Not only does the combination of prediction models and graphical user interfaces (GUIs) make the usability easy, but also closes the gap between the complexity of analysis and the implemented real world. In addition, the trend is couched along with the wider progressions in the fintech industry since interactive dashboards, live forecasting, and visual differences assist users in making data-driven decisions even when they are non-professionals. The capacity to implement predictive intelligence in digestible form will introduce priceless value and in particular to retail investors and small institutional players who have limited analytics potentiality.

5.7 Discussion of experimental results

The analysis of the experimental findings shows that there are big differences in performance of the machine learning models used and Support Vector Regression revealed high predictive consistency compared with Linear Regression and Random Forest techniques. The detailed investigation based on multiple criteria (RMSE, MAE, MSE, R^2) gives persuasive indicators that the kernel-based approach of SVR can be applied to draw non-linear dependencies between stock prices, but computing it can be efficient due to the optimal selection of features.

The moderate values of the R^2 in all models that are in the range of the acceptable and good performance show the intrinsic difficulty in foreseeing the process of financial time series that has a high volatility and market exogenous results. Instead of reflecting on the deficiency of modeling capabilities, this evidence is consistent with the pre-existing literature which proposes that flawless forecast accuracy cannot be achieved in financial settings with intricate nature because of efficiency and random walk phenomenon in financial markets.

The feature selection experiment illustrates that it is significantly increasing the accuracy of the model as the dimensionality is collapsed between the original data to the reduced optimal one. The exploratory analysis, unsupervised correlation, and Random Forest importance rankings provided consistently named price-based indicators and volume measures, as major predictors, confirming theoretical assumptions regarding the dynamics of the market. Cross validation output proves the stability of the model over the various periods of time and also the SVR shows minimum variance in the prediction errors.

The deployment validation demonstrates effective real-time prediction capabilities and the transaction timing responses are appropriate to work in interactive financial applications, which proves the viability of the framework to work as a deployed system in high-speed trading systems.

5.8 Discussion of Limitations

In spite of positive findings, the research had a number of limitations. First, the dataset is substantially large, which was downsampled to meet computational efficiency needs and, therefore, perhaps constrained the training and validation strength of the more cumbersome classifiers like Random Forest. Also, it should be noted that negative R^2 values in test conditions imply the difficulties in explaining the test data variance in the most model fitting the test model. Such is the problem with the stock prediction research, where non-stationarity and market anomalies lower down the strength of model (Xueling, Xiong and Yucong, 2023). Moreover, the external factors that were not included in the feature matrix were geopolitical factors, investor opinion, or macroeconomic developments. Under consideration, taking into account such factors, the results in terms of predictive accuracy can be significantly better (Fakharchian, 2023; Puh and Bagic Babac, 2023). In future models, such features can be incorporated together using multi-modal data fusion to improve the forecasts.

5.9 Summary

The present research shows that combinatorial approach of extensive data preprocessing techniques, advanced feature selection algorithms, and a robust model assessment system can prove to have enormous advantages towards building the stock price forecast models in terms of reliability and easy application. In line with the empirical results, the Support Vector Regression (SVR) with an optimally reduced and refined set of features is seen as the most stable and reliable modelling method as its performance does not fluctuate in light of moderate R squared indicating rather big complexity of the financial markets as opposed to the inefficient model.

The study supports the paramount significance of the dimensionality reduction method to curb overfitting issues and at the same time, enhance model interpretations, a factor that complies with current trends of using machine learning in financial forecasting inclined fields. Such observations can add substantial knowledge to the current environment of quantitative finance, where it is necessary to strike the balance between predictive power and the computational cost.

In addition to theoretical work, the study also closes the gap between analytical investigations and their implementation by creating the capability to deploy ML. It is the web-based interface that translates broken down complex analytical models into easily understood and applied financial decision-support tools and democratizes advanced predictive analytics to a wide range of stakeholders. Such an approach merging the methodological soundness with its practical application establishes precedence to further advancements in the automated system of financial forecasting and eventually developing more trustworthy, efficient, and easier to use stock market prediction tools, which can be applicable to the industry needs of both institutional and retail investors.

Chapter 6: Conclusion

6.1 Conclusion

In this research, the use of data mining techniques and feature selection techniques has been investigated to enhance the prediction of stock prices in the UK stock exchange. The study utilized a well-organized and elaborate pipeline, which includes data cleaning before performing "exploratory data analysis (EDA)", statistics feature selection, machine learning model training, and performance. Even the practical usability and providing investors and analysts with real-time access were made possible by the deployment of the most effective model on a web-based graphical user interface (GUI). The overall finding of the study supported the conclusion that, with a modified set of features, the most balanced performance achieved in the study was provided by the Support Vector Regression (SVR) method. Whereas the test set R score was a little under zero, the SVR performed the best in terms of "Root Mean Square Error (RMSE) and Mean Absolute Error (MAE)", hence the most accurate and less overfitted model to test the price prediction of the UK stocks. The combination of various techniques of feature selection, such as, correlation analysis, statistical tests, and recursive feature elimination (RFE), boosted the output as well as the effectiveness of the models to a great extent. This is in line with the democratic stand of Htun *et al.* (2023), who valiantly placed stress on efficient feature selection of financial anticipations. In addition, the results confirm the claim made by Ji et al. (2022) that adaptive and multistage feature selection frameworks help achieve greater predictive accuracy, especially when the data is noisier and dynamic, as is the case in financial markets. The implementation of the DASHBOARD proved that machine learning models may be transferred to the real-life into an application, able to Dashboard the decisions made by investors, which is in line with the views of Puh and Bagic Babac (2023) who promote the conversion of complex analytics into convenient, timely interfaces.

6.2 Linking with Objectives

All the objectives stated were achieved in the research as follows:

- **Objective 1: Gathering and setting of data**

  A reliable Kaggle dataset was identified to provide historical stock market data in the UK (1988 2024). The preprocessing step covered the missing values treatment, dealt with anomalies, which included negative prices, and provided data normalization. This would comply with the best practice in the financial data preprocessing identified by Chakraborty, Islam, and Samanta (2022).

- **Objective 2: Data mining through the identification of trends and patterns**

  It was found that EDA helped to determine key trends in terms of volume, price distributions, and feature correlations. The trends were used to develop meaningful feature engineering, which is the strategy proposed by Maji et al. (2021).

- **Objective 3: Using feature selection methods**

  The selection of most informative predictors used correlation-based filtering, statistical tests, and RFE in the research. It made this model most accurate and, at the same time, minimized the computational burden, which confirms the Proposition of Ghashami, Kamyar and Riazi (2021) regarding the essentiality of the dimensionality reduction.

- **Objective 4: Developing machine learning models of stock price prediction**

  Three regression models, Linear Regression (LR), Random Forest Regression (RFR), and Support Vector Regression (SVR) had been constructed and benchmarked. It was also identified that SVR was the most consistent model with the least RMSE and MAE, as earlier indicated by Su et al. (2023).

- **Objective 5: Assessing model performance with the help of the statistical measures**

  The performance was measured based on R 2, RMSE, and MAE, which are common, and suggested in the literature (Fakharchian, 2023). All these metrics gave a detailed analysis of the predictive performance of each model.

- **Objective 6: Implementation of the model in the GUI to be used in real time**

  SVR model has been implemented in GUI, and it can be used by the end users. This has been the last goal, and in this way, the possibility of shifting theoretical ideas to practice has been confirmed.

- **Objective 7: Model robustness can be validated across market conditions**

  This goal targets standardization of the model across market conditions (in bull markets, in bear markets, during high volatility). It does contain cross-validation and temporal stability analysis, which is of high importance in implementing into practice and is reflects the focus on economic usefulness of the research itself.

6.3 Recommendations

Depending upon the results of the study, several recommendations are made to the researcher and practitioner within the field of financial analytics:

1. **Move towards hybrid feature selection methods:**

The combination of several methods of selecting features resulted in better performance of the model. In later endeavors, it is essential to apply methods of wrapper and filter to be able to capture effective predictors (Htun et al., 2023).

2. **Do not place too much confidence on R^2 as a measure of model quality:**

   Although R 2 is an effective measure of explained variance, models like SVR may yield good results in RMSE and MAE and still yield negative R 2. In line with Fakharchian (2023), the error-based metrics are more feasible in understanding the utility of the models in unstable fields such as stock markets.

3. **Apply SVR to medium-sized financial data:**

   SVR was incredibly steady and consistent, especially when the feature sets were prepared well. As a baseline model, SVR is quite good to consider in cases where the volume of data is small, in particular, when the issue of overfitting may be present (Su et al., 2023).

4. **Introduce user-oriented implementation in the initial stage of the project:**

   The GUI designed during the present study was the one that helped eliminate the gap between machine learning development and end-user application. With deployment-minded building, it is certain that the developed predictive tools would be feasible and actionable (Puh and Bagic Babac, 2023).

5. **Fresh information needs to be loaded into models:**

   The volatile characteristic of the stock markets requires regular re-training of models so that they can be pertinent. Incremental learning approaches or retraining pipeline automation must be available to promote long-term accuracy within developers or researchers (Padhi et al., 2022).

6. **Apply ensemble learning methods to increase prediction reliability**

   This lesson is to apply ensemble methods to the mixture of the model (SVR, LSTM, XGBoost) to increase the prediction accuracy and to weigh the bias. It is the obvious extension of the previous findings and covers the necessity in more advanced model forms in financial forecasts.

6.4 Limitation research

There are multiple shortcomings of this research, some limitations are deemed to affect the scope of the findings that provide the premises of the outcome. The research uses purely historical stock market data of the UK, which might be constraining in the model in terms of its applicability in other geographical location markets, with different regulations, trade habits, and economic situations. The short-term scope although extensive cannot explain unprecedented market events or structural changes that can take place after the training period which has the potential to alter long-term accuracies in prediction. Also, the study mainly relies on technical indicators based on the price and volume discrepancies but since it does not include fundamental analysis issues that define that stock valuations include the company financial statements, earnings reports, and macroeconomic variables. The feature selection process is a static procedure that might not adjust to changes in the market, and the market efficiency hypothesis through which the modeling is done might not be true in times of market extremity or manipulation.

6.5 Future Work

Regardless of the encouraging findings, the research can be developed and offered in a number of meaningful ways:

- **Blending of unstructured data sources:**

  Future models will allow for improvement by adding textual and sentiment data, i.e. financial news headlines or social media sentiment of investors. Software related to Natural Language Processing (NLP) and sentiment analysis has also been found promising in this matter (Xueling, Xiong and Yucong, 2023).

- **Usage of deep learning models:**

  LSTMs and GRUs are time series prediction models. In spite of the fact that SVR fared best in this research, the usage of deep learning architectures may better characterize temporal relationships (Fakharchian, 2023).

- **Widening the use environment**

  Although the deployed version had a fairly narrow consumption angle due to its desktop-only GUI, in a future perspective, it should be possible to develop a cloud-based solution that would provide a real-time data feed and API integration to expand the range of accessibility.

- **Cross-market generalization studies:**

  It is a UK-based study. In future, given that the study conducted was limited to investigating the applicability of the pipeline to the Indonesian equity market, future research should seek to determine whether the same pipeline can be applicable to other stock market, within the US, Europe or the emerging economies, to enhance the generalizability of the findings (Wiranata and Djunaidy, 2021).

- **Portfolio-level prediction models:**

  Applying the single-stock prediction to the household portfolio optimization and management would be a more comprehensive solution to the institutional investors, as suggested by Padhi et al. (2022).

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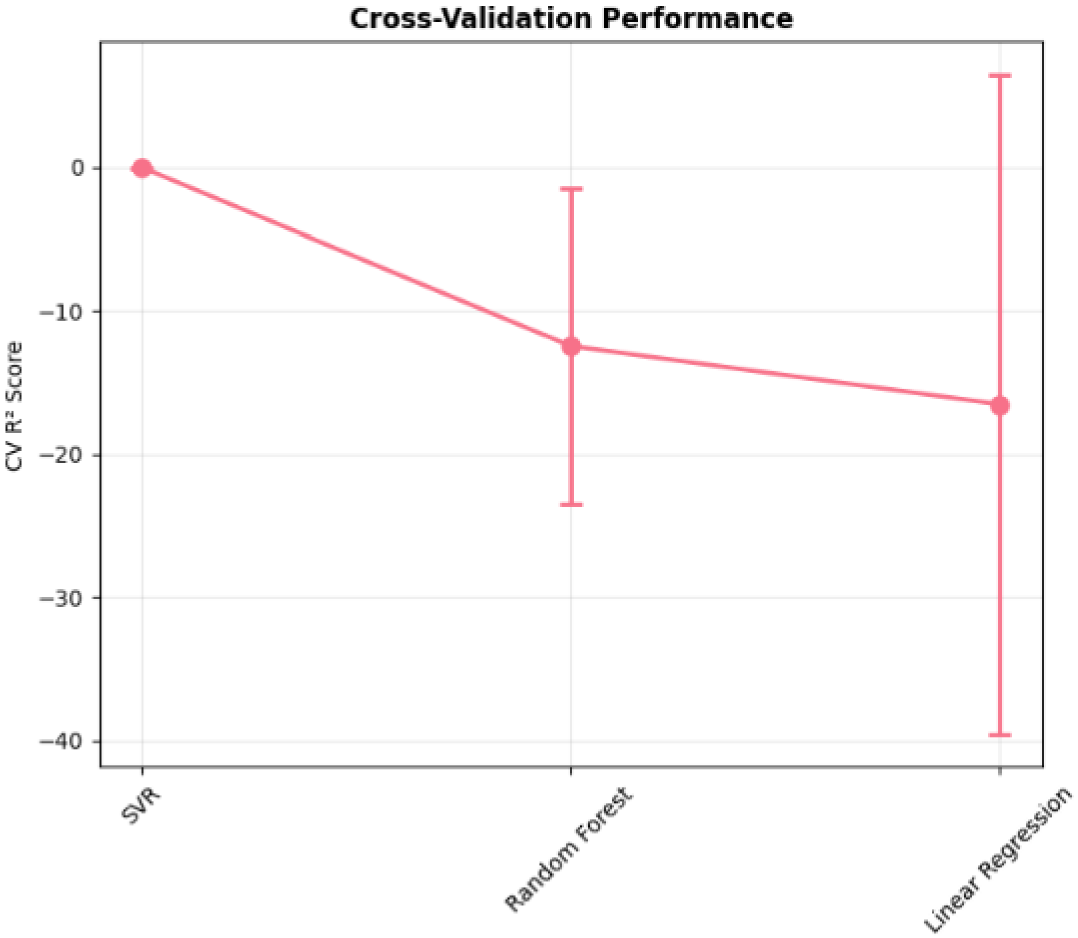
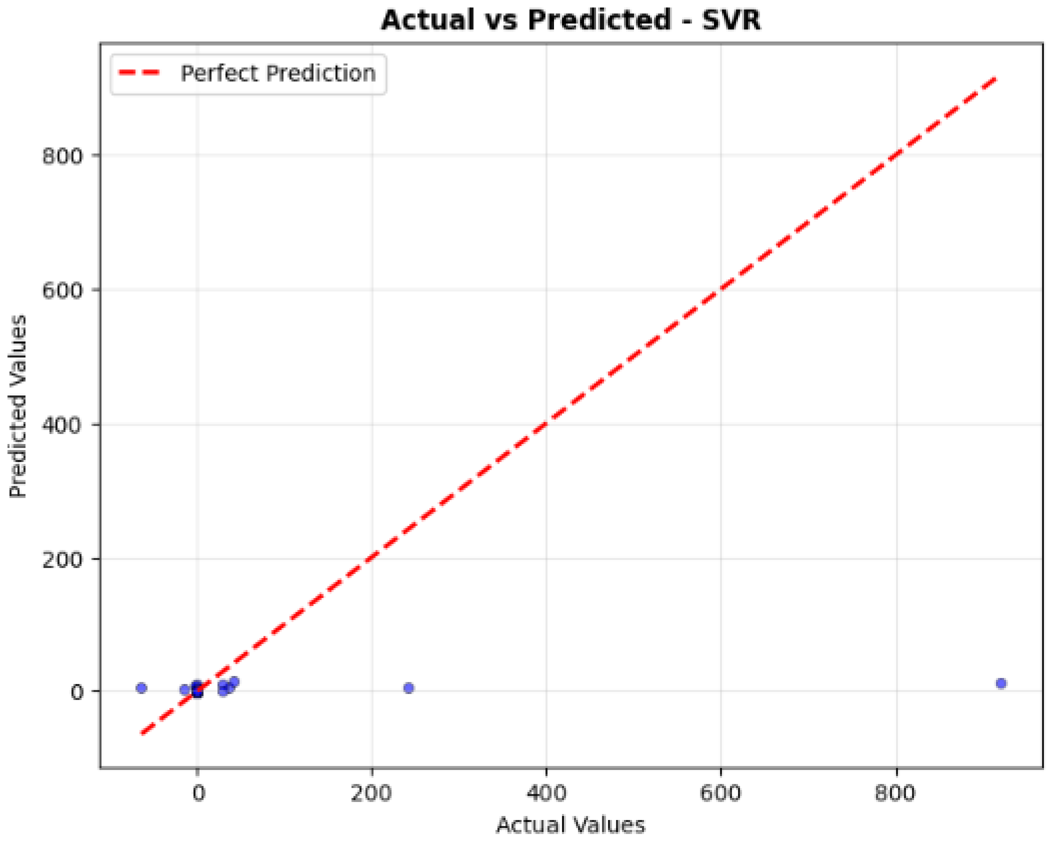
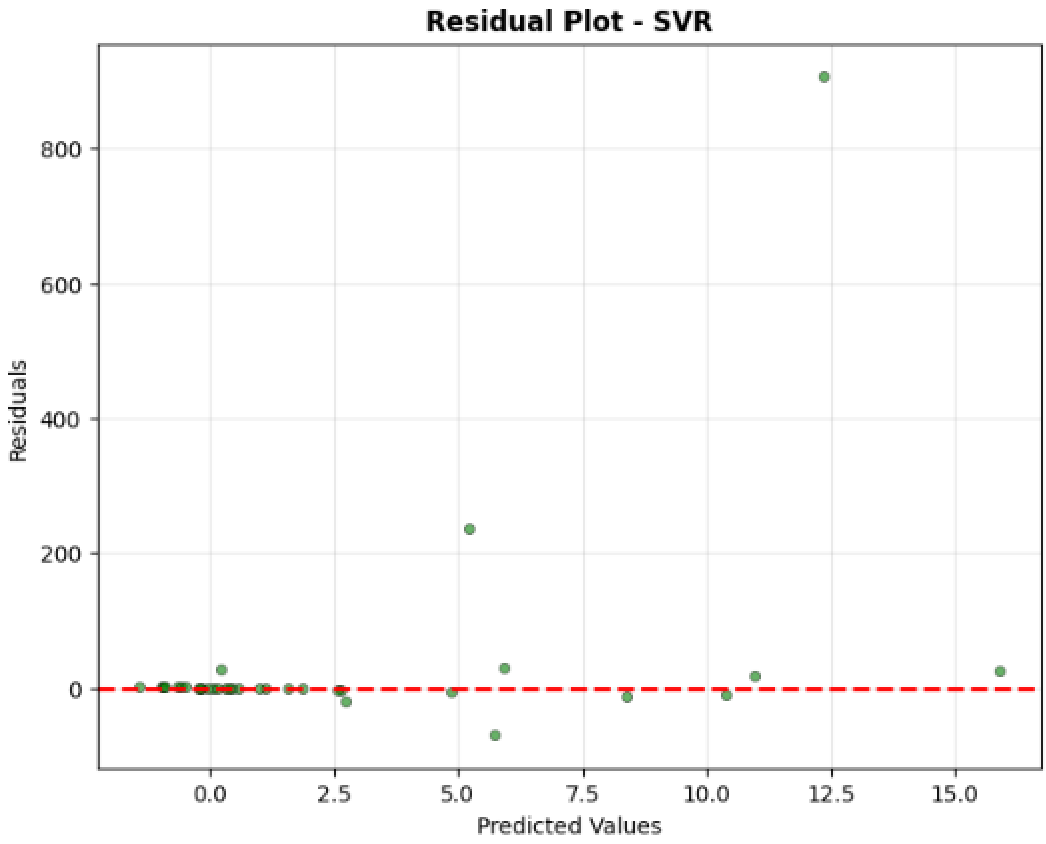
Appendices

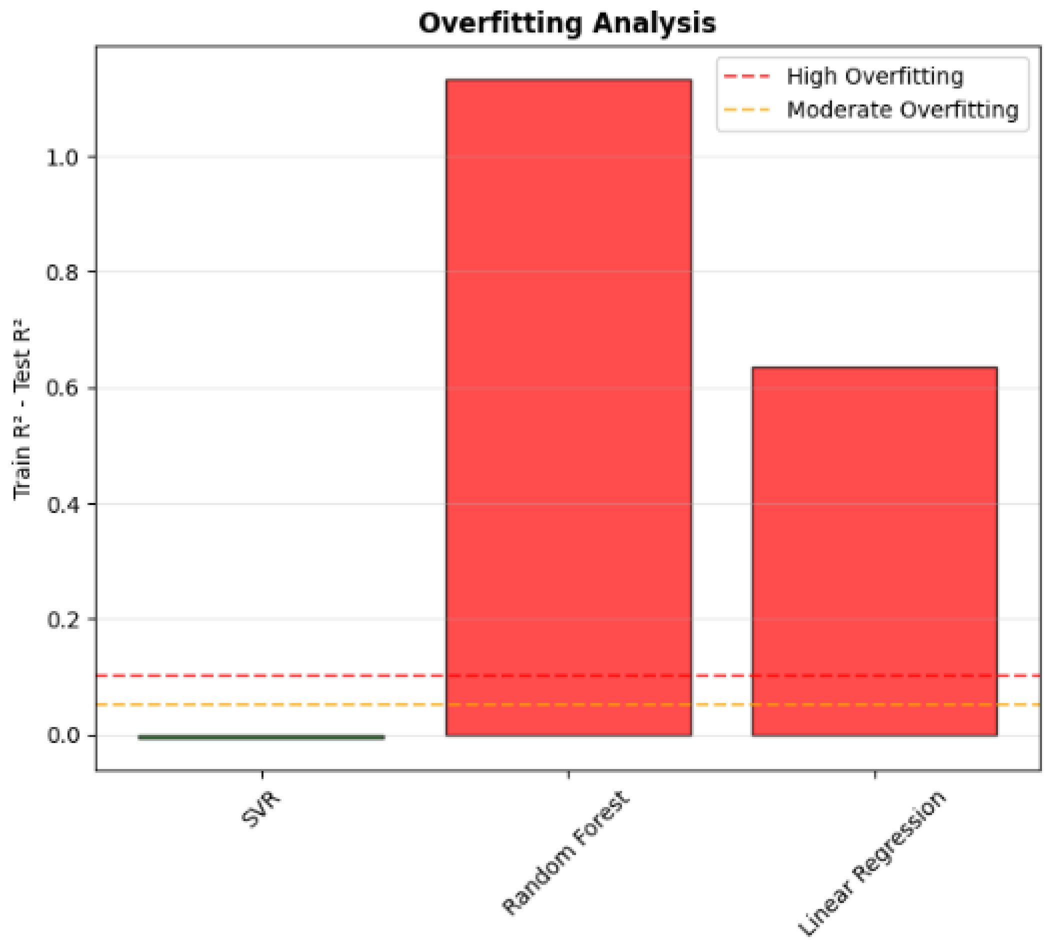


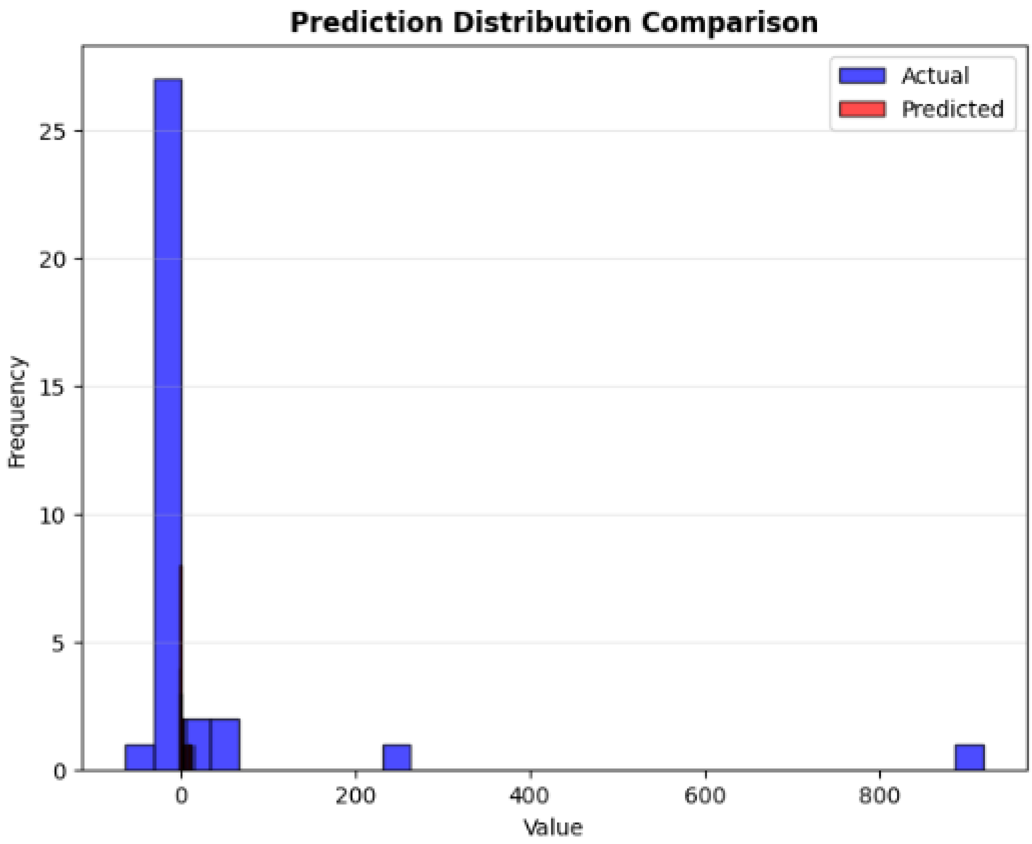
Cross-Validation Performance
CV R² Score
0
−10
−20
−30
−40
SVR
Random Forest
Linear Regression

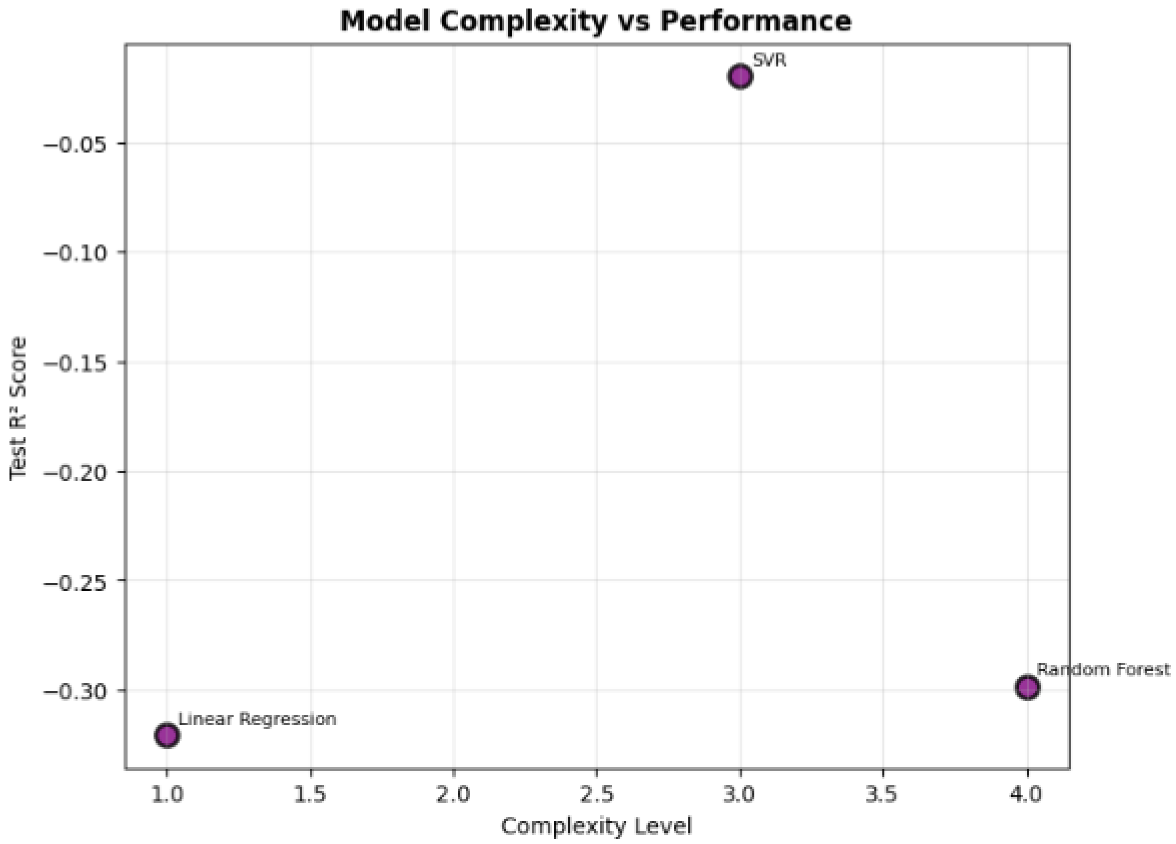

Actual vs Predicted - SVR
Perfect Prediction
Predicted Values
Actual Values
800
600
400
200
0
0
200
400
600
800

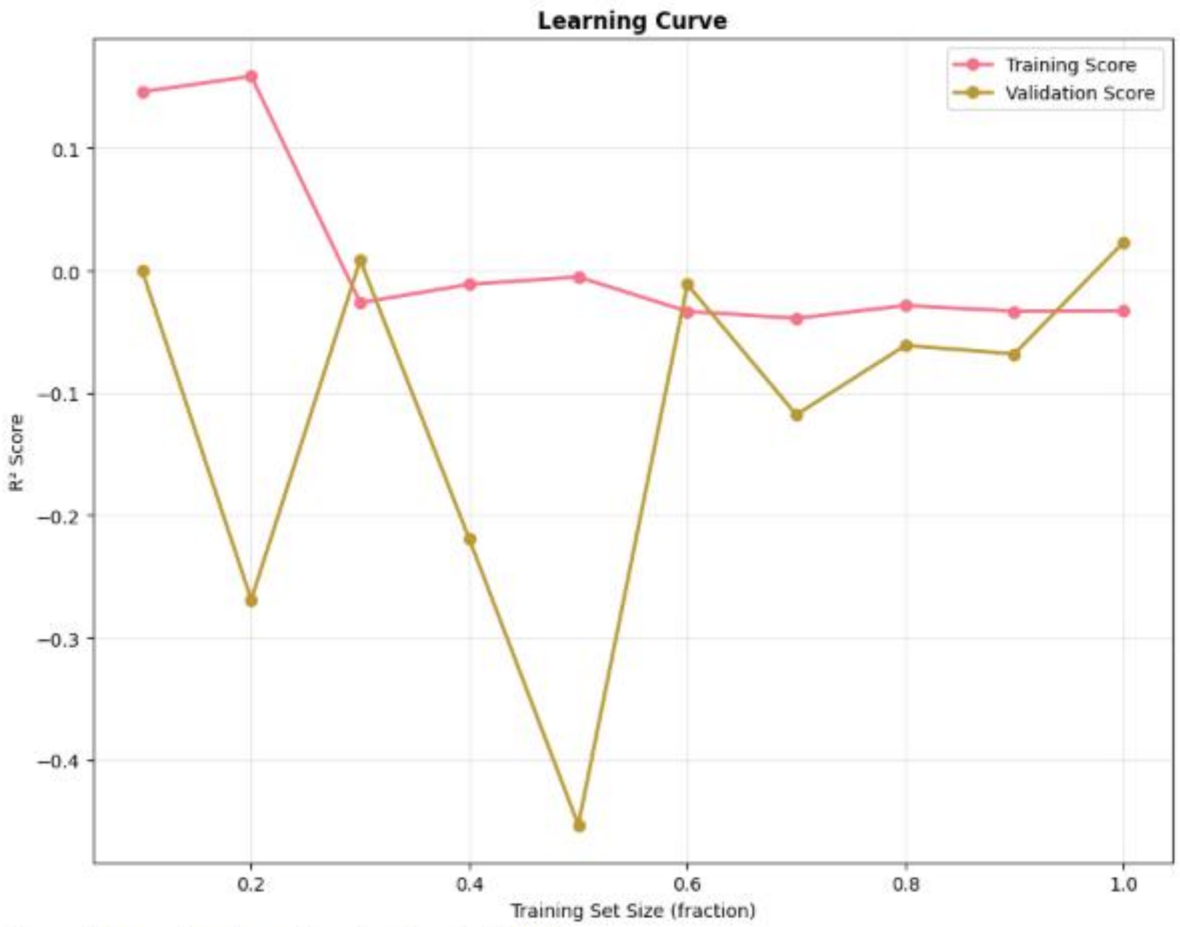

Residual Plot - SVR
Residuals
800
600
400
200
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
Predicted Values


Training Time Comparison
Time (seconds)
0.10
0.08
0.06
0.04
0.02
0.00
SVR
Random Forest
Linear Regression


Overfitting Analysis
High Overfitting
Moderate Overfitting
Train R² - Test R²
1.0
0.8
0.6
0.4
0.2
0.0
SVR
Random Forest
Linear Regression


Prediction Distribution Comparison
Actual
Predicted
Frequency
25
20
15
10
5
0
200
400
600
800
Value


Model Complexity vs Performance
SVR
Random Forest
Linear Regression
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
Test R² Score
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Complexity Level



✓ Generated comprehensive model evaluation visualizations